

Examples of Types of Stem Plots

Example: Test Scores

Suppose the test scores for 24 students are:

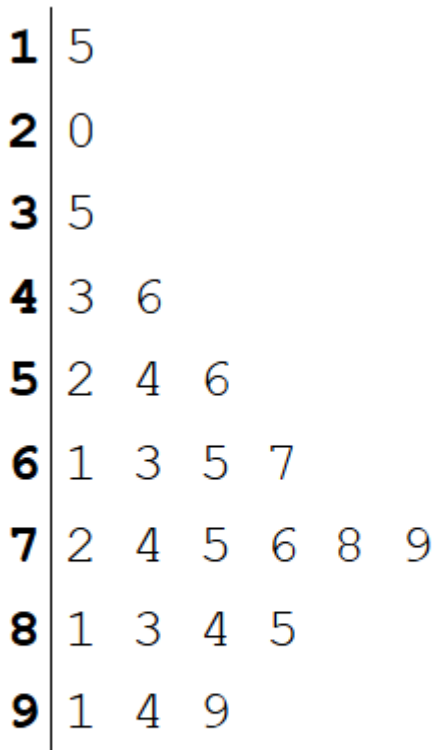
Data Set A:

15, 20, 35, 43, 46, 52, 54, 56, 61, 63, 65, 67, 72, 74, 76, 78, 79, 81, 83, 84, 85, 91, 94, 99

1-Simple Stem Plot:

In a simple stem plot, each stem represents the tens digit, and the leaf represents the ones digit.

Key: 6 | 5 = 65



Test Scores of Class A

KEY: 9 | 9 = 99

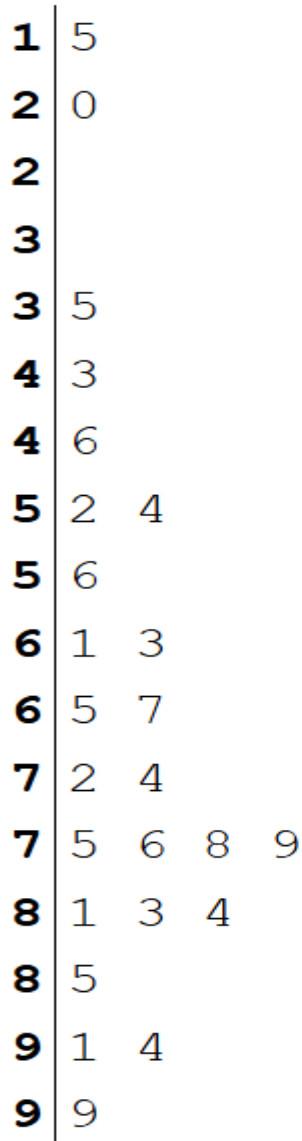
2. Split Stem Plot:

The problem with the simple stem plot is that there are many leaves on each stem. A split stem plot gives us more detail.

Each stem is split into two rows:

First row: leaves 0–4

Second row: leaves 5–9



Why split the stems?

Instead of putting all the scores in the 60s in one row:

6 | 1 3 5 7

we can see the distribution more clearly:

6 | 1 3

6 | 5 7

Test Scores of Class A

KEY: 9 | 9 = 99

3-Compare Two Data Sets with a Side-By-Side Stem Plot:

Now suppose we have scores from **Class A** and **Class B**.

Class A

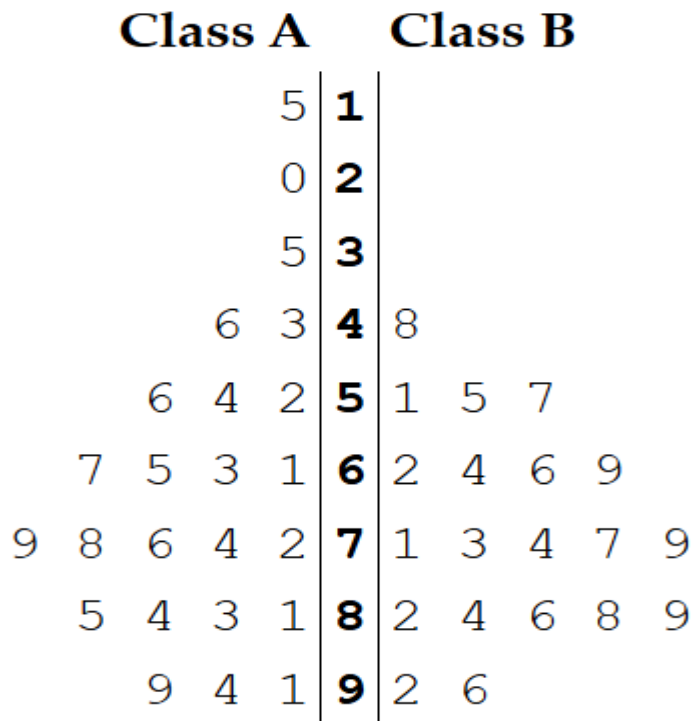
15, 20, 35, 43, 46, 52, 54, 56, 61, 63, 65, 67, 72, 74, 76, 78, 79, 81, 83, 84, 85, 91, 94, 99

Class B

48, 51, 55, 57, 62, 64, 66, 69, 71, 73, 74, 77, 79, 82, 84, 86, 88, 89, 92, 96

We can display both sets using a **side-by-side stem plot**:

Notice that the leaves for Class A are read **from right to left**, while the leaves for Class B are read **from left to right**.



Test Scores

KEY: 9|9 = 99

How to Write Perfect SOCS:

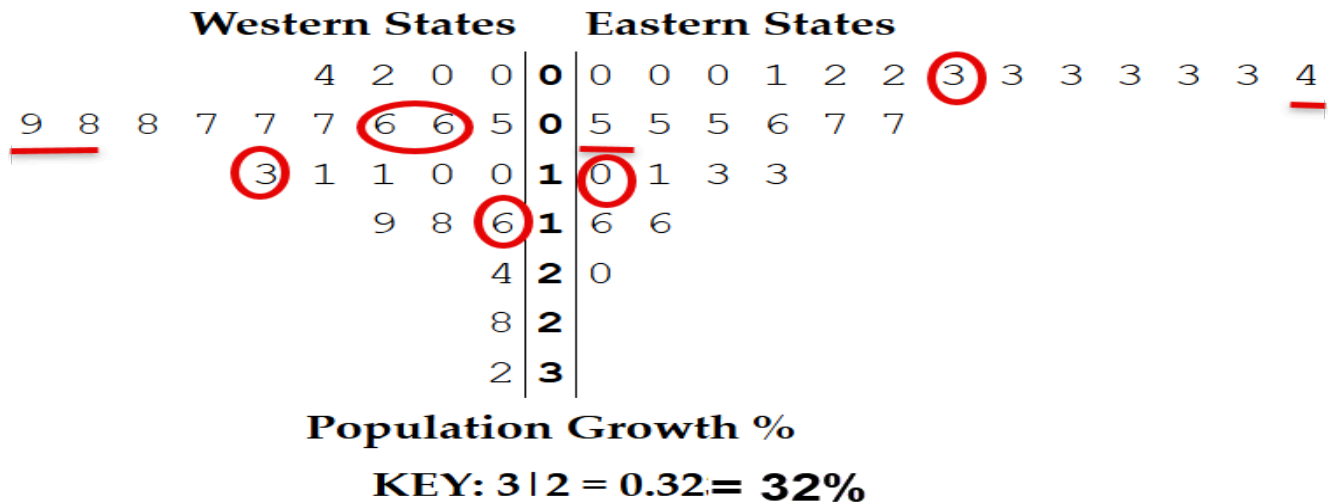
1. **Context (C):** Always start by stating what the data represents and specifying the variables and units.
2. **Shape (S):** Describe the visual structure—whether it is symmetric, skewed (left/right), or unimodal/bimodal/uniform.
3. **Outliers (O):** Check for extreme values or unusual observations (using rules like $1.5 \times \text{IQR}$).
4. **Center (C):** State the typical value—use **median** for skewed data/outliers or **mean** for symmetric distributions.
5. **Spread (S):** Describe the variability—use **IQR** alongside median, or **Standard Deviation** alongside mean.
6. **Conclusion (C):** Tie all the statistical evidence back together in full sentences to summarize what it means in context.

Key Takeaways for Writing Your Paragraph

- **Use Full Sentences:** Avoid bullet points when writing your formal response.
- **Always Include Units:** Don't just list numbers (e.g., say "*a median of 45 miles per hour*" rather than "*a median of 45*").
- **Pair Correctly:** Always pair **Median with IQR** and **Mean with Standard Deviation**.

Activities 7-4, 8-15 and 10-10: Population Growth

Reading Stemplot



Finding the median from the stemplot:

There are 24 western states, so the median is the average of the data values in the middle at positions 12 and 13

$$\text{Median} = (8+9)/2 = 8.5\%$$

There are 26 eastern states, so the median is the average of the data values in the middle at positions 13 and 14

$$\text{Median} = (4+5)/2 = 4.5\%$$

When trying to locate the data position, start from the smallest stem and go over the leaves from inside to outside (start at the stem and count out)

Finding the quartiles from the stemplot:

Western: Q1 is the average of the data values at positions 6 and 7 . $Q1 = (6+6)/2 = 6\%$

Q3 is the average of the data values at positions 18 and 19 . $Q3 = (13+16)/2 = 14.5\%$

Minimum = 0%

Maximum = 32%

Eastern: Q1 is the data value at position 7 . $Q1 = 3\%$

Q3 is the data value at position 7 from the end or 20 from the beginning $Q3 = 10\%$

Minimum = 0%

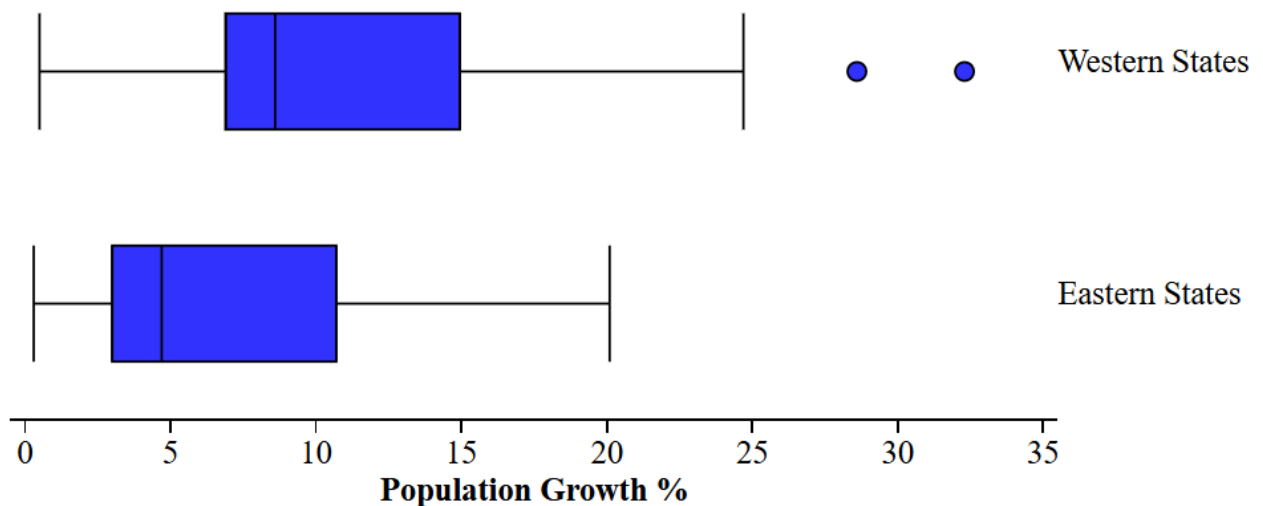
Maximum = 20%

Values in the table are in percentage

Group	Name	Min	Q1	Median	Q3	Max
1	Western	0	6	8.5	14.5	32
2	Eastern	0	3	4.5	10	20

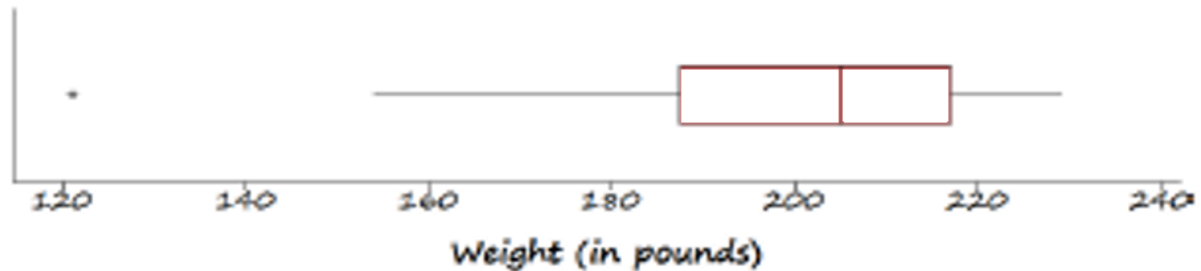
Outlier Calculations

<u>Western:</u>	<u>Eastern:</u>
$\begin{aligned} \text{IQR} &= Q3 - Q1 \\ &= 14.5 - 6 \\ &= 8.5 \end{aligned}$	$\begin{aligned} \text{IQR} &= Q3 - Q1 \\ &= 10 - 3 \\ &= 7 \end{aligned}$
$\begin{aligned} 1.5 \text{ IQR} &= 1.5(8.5) \\ &= 12.75 \end{aligned}$	$\begin{aligned} 1.5 \text{ IQR} &= 1.5(7) \\ &= 10.5 \end{aligned}$
$\begin{aligned} \text{LF} &= Q1 - 1.5 \text{ IQR} \\ &= 6 - 12.75 \\ &= -6.75 \end{aligned}$	$\begin{aligned} \text{LF} &= Q1 - 1.5 \text{ IQR} \\ &= 3 - 10.5 \\ &= -7.5 \end{aligned}$
No lower outliers	No lower outliers
$\begin{aligned} \text{UF} &= Q3 + 1.5 \text{ IQR} \\ &= 14.5 + 12.75 \\ &= 27.25 \end{aligned}$	$\begin{aligned} \text{UF} &= Q3 + 1.5 \text{ IQR} \\ &= 10 + 10.5 \\ &= 20.5 \end{aligned}$
There are two upper outliers a 28% and 32%	No upper outliers
maximum data value that is not outlier is 24%	
Right whisker will extend to 24%	



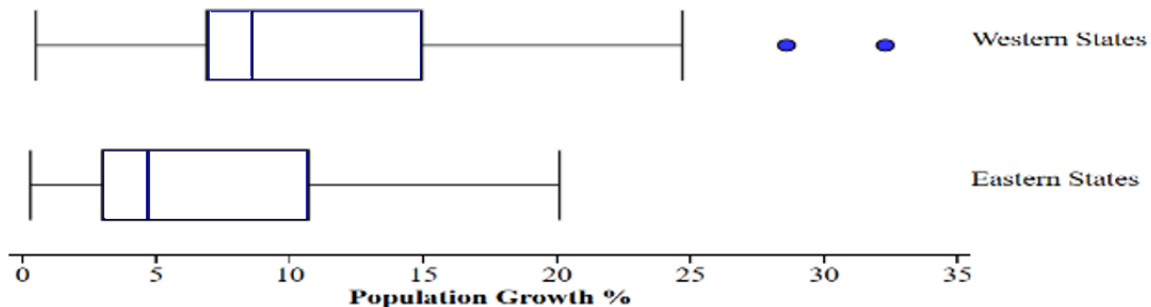
SOCS Examples V-1

Writing a SCOS describing the features of a distribution (one group)



The distribution of the weights of the rowers on the men's 2008 Olympic team in pounds is left-skewed. Based on the calculations $(188 - 1.5 \cdot 30, 218 + 1.5 \cdot 30)$, there is one outlier. The median is 205 lbs., and the IQR is 30 lbs. A typical rower tends to weight 205 lbs. and there is some variability.

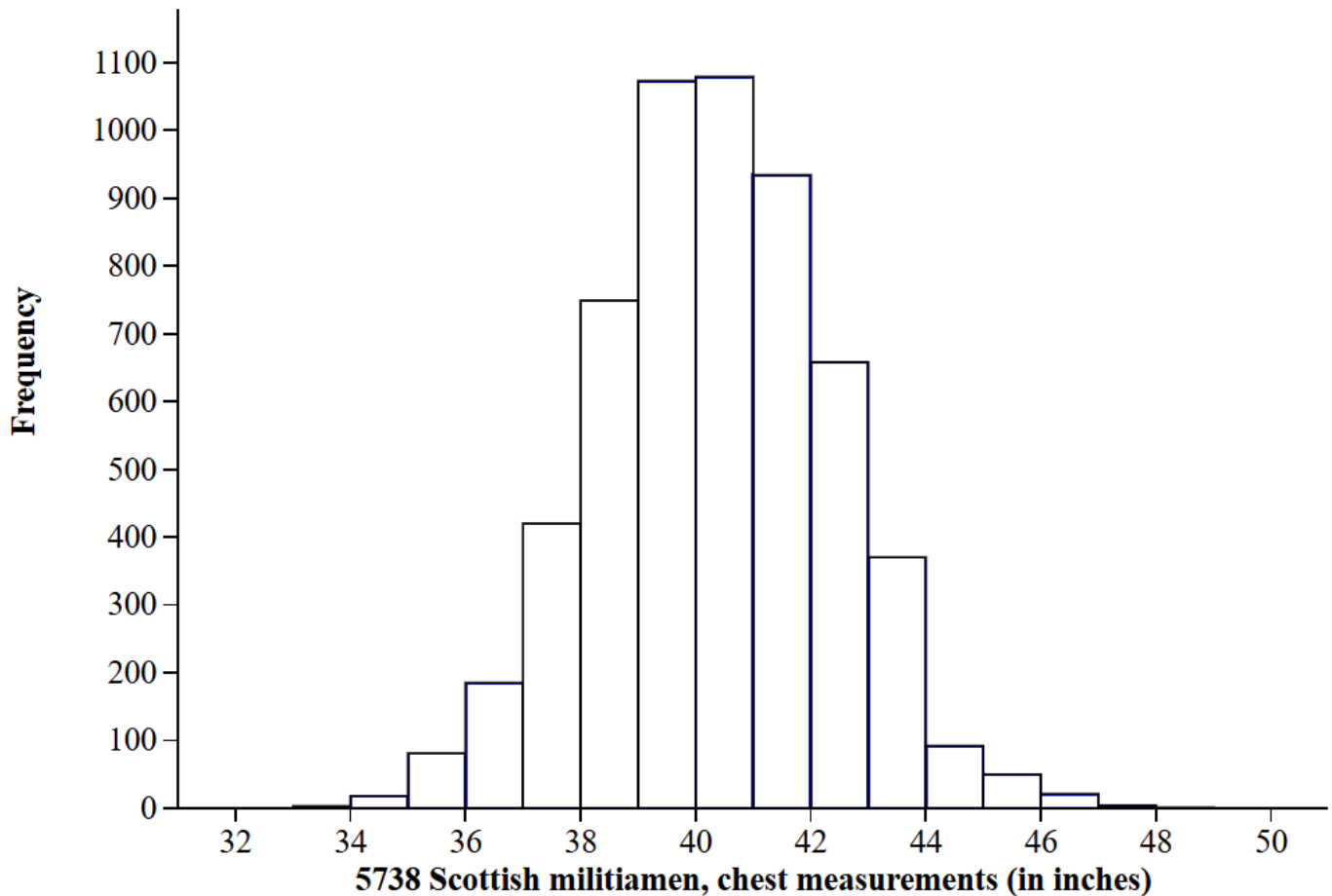
SOCS of two distributions (both have the same shape)



The distributions of percentage of population growth for both the western and eastern states are skewed to the right. Based on the calculations $(7 - 1.5 \cdot 8, 15 + 1.5 \cdot 8)$ there are outliers in the western states, and based on the calculations $(3 - 1.5 \cdot 8, 11 + 1.5 \cdot 7.7)$, there are no outliers in the eastern states. The median of the western states is 9% and the median of the eastern states is 5%. The IQR of the western states is 8% and the IQR of the eastern states is 7.5%. A typical western state tends to have higher percentage population growth rates than do the eastern state, but there is more variability in population growth in the western states.

SOCS Examples V-2

Writing a SCOS describing the features of a distribution (one group):



Summary Statistics

n	mean	SD	min	Q ₁	med	Q ₃	max
5738	39.832	2.05	33	38	40	41	48

The distribution of the chest sizes of 5738 Scottish militia men in **inches** is mostly symmetric. Based on the calculations **(38-1.5*3, 41+1.5*3)**, there are outliers. The median is **40 inches**, and the IQR is **3 inches**. A typical Scottish militiaman **tends to** have a chest size of **40 inches** and there is **some variability**.

SOCS of two distributions (They have different shapes):

First Class										Last Class													
2										5	4	7	8										
6 0										6	0	6	6	6	8								
6	4	4	2	0						7	0	1	2	2	3	4	4	4	5	6	6	7	
6	6	4	2	2	2	2	0	0						8	0	0	2	2	2	5	6		
8 2 2 0										9	2	5	6										

AP Statistics Tests Scores (in percentages)

KEY: 9|8 = 98

Summary Statistics

Group Name	n	mean	SD	min	Q ₁	med	Q ₃	max
1: First Class	21	79.048	11.02	52	73	82	86	98
2: Last Class	30	74.633	10.483	54	68	74	82	96

The distribution of the AP Statistics test scores **in percentage** for the First class is left-skewed whereas for the Last class is roughly symmetric. Based on the calculations (**73-1.5*13, 86+1.5*13**), there is one outlier for the first class, and based on the calculations (**68-1.5*14, 82+1.5*14**), there are no outliers for the Last class. The median of the AP Statistics Tests Scores for the First class is **82%** and the median of AP Statistics Tests Scores for the Last class is **74%**. The IQR of the AP Statistics Tests Scores for the First class is **13%** and the IQR of the AP Statistics Tests Scores for the Last class is **14%**. A **typical** AP Statistics Test Score in percentages from the First class **tends to be higher than** this from the Last class and the two classes have **similar variability**.

SOCS Review

Features of a Distribution:

Center (Mean, Median and Mode are measures of the center)

Spread (Range, Standard deviation and IQR are measures of the spread)

Shape (symmetric, skewed left, skewed right,

Outliers (outliers calculation and their effects on the measures of center and spread)

The Mechanics Behind Writing a C-SOCS-C

Context:

- Observational units
- Variable
- Measuring units
- Name of the groups

Shape

- Given graphs (where is the tale located)
- Given data Table with frequencies
(where less frequencies located is the tale of the distribution)
- Comparing Mean and Median from summary statistics

Outlier

- Outliers calculation
- Read the data specifically form dot plot or stem plot

Center and Spread

- Use the right measure
- If symmetric and no outliers use Mean, SD
- If skewed and/or there are outliers use median and IQR

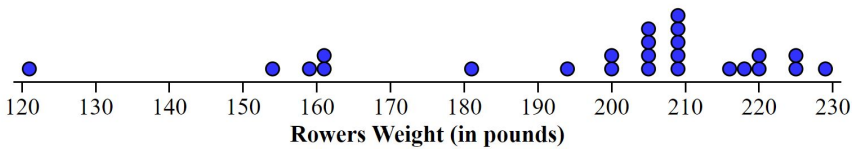
Conclusion

- If one group - only use number related to the center and low variability or high variability
- If two groups compared - only words and no numbers to compare the typical value and variability

In C-SOCS-C make sure that:

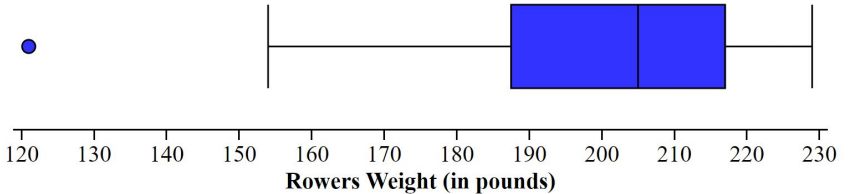
- units of measurements included after each numerical value
- All the items are there C-SOCS-C
- Do the outliers calculations (LF, UF) aside:
 - if box plot is given look to see if there are any outliers
 - if dot plot or stem plot is given look for the smallest and largest data points and compare their values to (LF, UF) to see if there are any outliers
 - if histogram is given look to see if there is any bin far away from most of bins and look for gaps before it.
- Make sure correct measures are used and their values mentioned
- IQR or SD has to be only single numerical value. No Range please
- Conclusion has to mention only the value of the measure of the center and that there is a lot or less variability for single distribution
- Conclusion has to compare the measures of the centers and spreads without using and numerical values when comparing two distributions

The distribution of the weights of the rowers on the men's 2008 Olympic team and the summary statistics are given, write a SOCS describing this distribution.

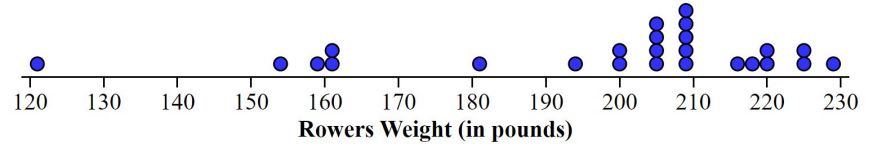


Summary Statistics

n	mean	SD	min	Q ₁	med	Q ₃	max
25	197.96	26.886	121	187.5	205	217	229

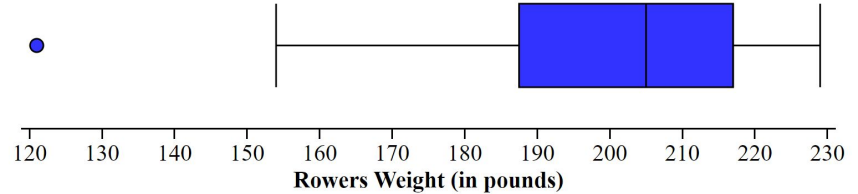


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Summary Statistics

n	mean	SD	min	Q ₁	med	Q ₃	max
25	197.96	26.886	121	187.5	205	217	229

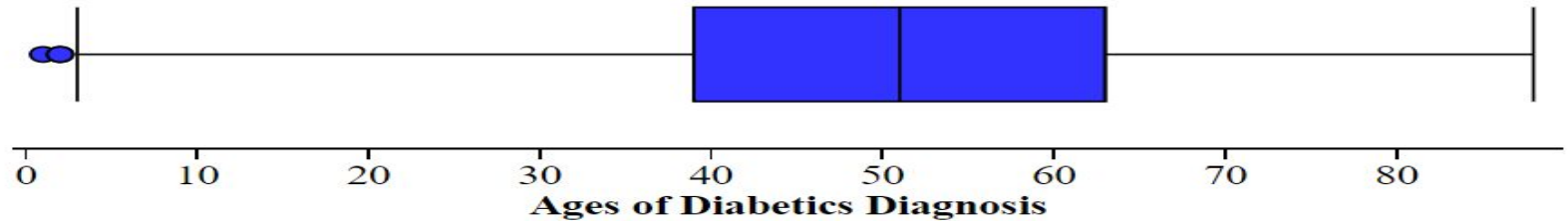


Answer:

The distribution of the weights of the rowers on the men's 2008 Olympic team in pounds is left-skewed. Based on the calculations $(188 - 1.5 \cdot 30, 218 + 1.5 \cdot 30)$, there is one outlier. The median is 205 lbs., and the IQR is 30 lbs. A typical rower tends to weight 205 lbs. and there is some variability.

Find the mistakes in this SOCS and correct it:

The distribution of the ages of diabetics diagnosis among 548 people is in years right skewed. The mean is 49.281, and the IQR is 39 to 63 . Therefore, most ages of diabetics tend to be between 39 to 63 but one diagnosis at age one years old

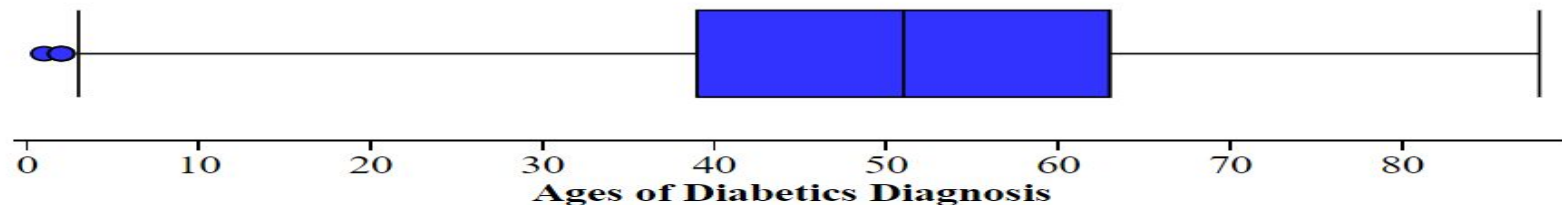


Summary Statistics

n	mean	SD	min	Q ₁	med	Q ₃	max
548	49.281	19.211	1	39	51	63	88

Answer:

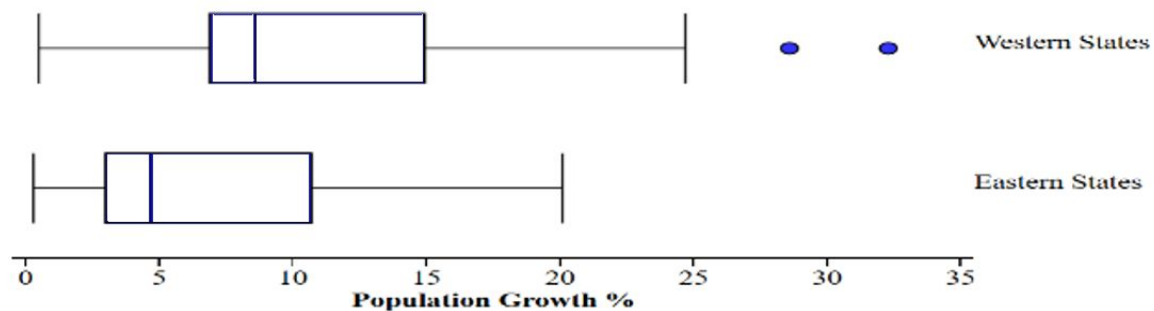
The distribution of the ages of diabetics diagnosis among 548 people in years is left skewed. Based on the calculations $(39 - 1.5 * 24, 63 + 1.5 * 24)$ there are outliers. The median age is 51 years old, and the IQR is 24 years. A typical diabetic person tends to be 51 years old and there is a lot of variability.



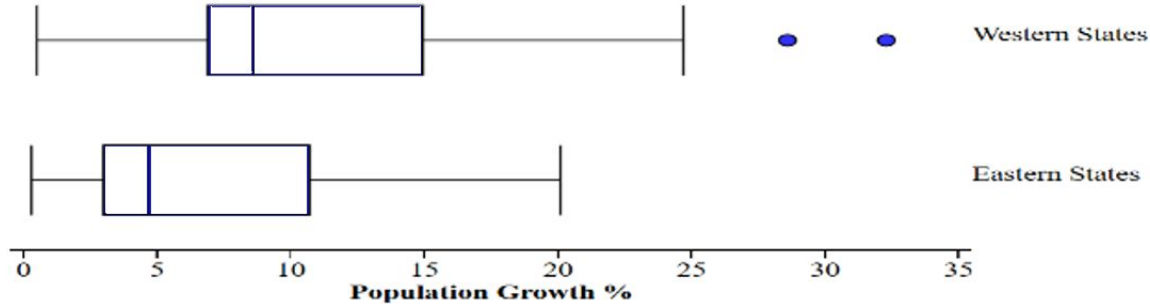
Summary Statistics

n	mean	SD	min	Q ₁	med	Q ₃	max
548	49.281	19.211	1	39	51	63	88

Write SOCS comparing the given distributions



Answer



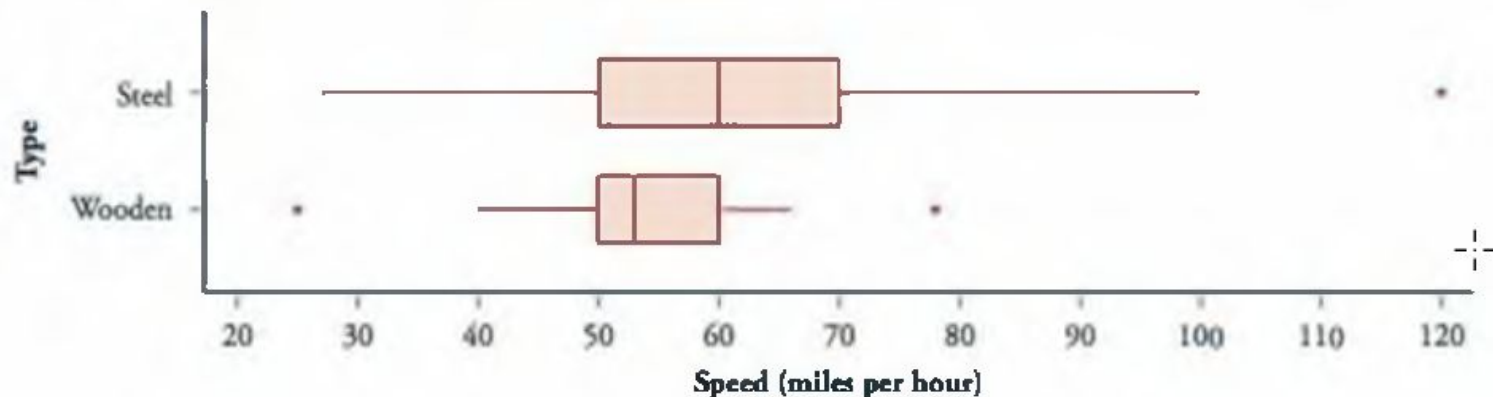
The distributions of percentage of population growth for both the western and eastern states are skewed to the right. Based on the calculations $(7 - 1.5 * 8, 15 + 1.5 * 8)$ there are outliers in the western states, and based on the calculations $(3 - 1.5 * 8, 11 + 1.5 * 7.7)$, there are no outliers in the eastern states. The median of the western states is 9% and the median of the eastern states is 5%. The IQR of the western states is 8% and the IQR of the eastern states is 7.5%. A typical western state tends to have higher percentage population growth rates than do the eastern state, but there is more variability in population growth in the western states.

Warm Up

Write a SOCS for comparing these two distribution

Activity 10-2: Roller Coasters 2-17, 10-2

Recall from Exercise 2-17 that the Roller Coaster DataBase is an online resource that contains data on hundreds of roller coasters around the world. The following boxplots display the distributions of top speeds (in miles per hour) for coasters in the United States, classified by whether the coaster is wooden or steel.



Answer:

The distribution of the top speed in of hundreds of roller coasters around the world of the Steel Type in miles per hour is symmetric, while the distribution of the top speed of hundreds of roller coasters around the world of the Wooden Type in miles per hour is right-skewed.

Based on the calculations ($50 - 1.5 * 20$, $70 + 1.5 * 20$) there is an outlier for the Steel Type roller coasters and based on the calculations ($50 - 1.5 * 10$, $60 + 1.5 * 10$) there are outliers for the Wooden Type roller coasters. The median of the Steel Type roller coasters is 60 miles per hour and the median of the Wooden Type roller coasters is 53 miles per hour. The IQR of the Steel Type roller coasters is 20 miles per hour and the IQR of the Wooden Type roller coasters is 10 miles per hour. A typical Steel Type roller coasters tends to go faster than a typical Wooden Type roller coasters and there is more variability with the Steel Type roller coasters than the Wooden type roller coasters

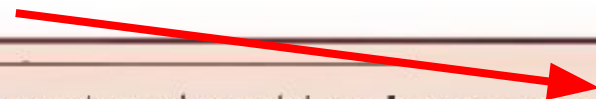
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Topic 4

From Topic 3:



A sampling procedure is said to have **sampling bias** if it tends systematically to overrepresent certain segments of the population and to underrepresent others.

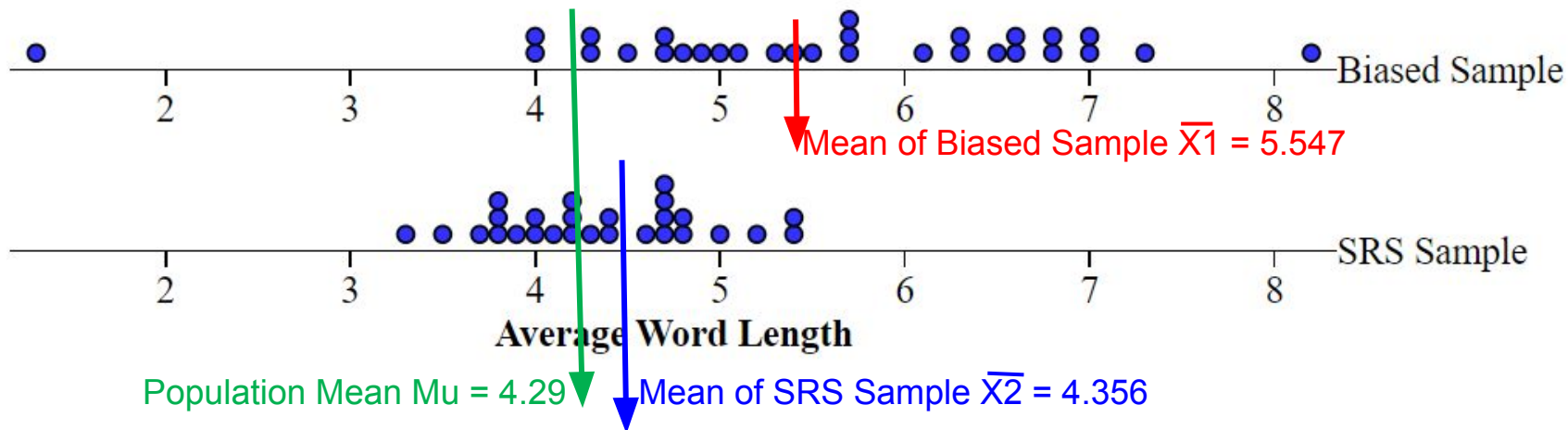
One way to avoid a biased sampling method is to give every member of the population **the same chance of being selected** for the sample. Moreover, your selection method should ensure that every possible sample (of the desired sample size) has an equal chance of being the sample ultimately selected. Such a sampling design is called **simple random sampling**.

A statistic is said to provide **unbiased** estimates of a population parameter if values of the statistic from different random **samples are centered at the actual parameter value**.

An important statistical property known as **sampling variability** refers to the fact that **the values of sample statistics vary from sample to sample.**

The **precision** of a sample statistic refers **to how much the values vary from sample to sample.** Precision is related to sample size: Sample statistics from larger samples are more precise and closer together than those from smaller samples. Statistics from larger random samples, therefore, provide a more *accurate* estimate of the corresponding population parameter.

Although the role of sample size is crucial to assessing how a sample statistic varies from sample to sample, the size of the population does not affect this sampling variability. As long as the population is large relative to the sample size (at least 10 times as large), the precision of a sample statistic depends on the sample size, not on the population size.



Summary Statistics

Group Name	n	mean	SD	min	Q ₁	med	Q ₃	max
1: Biased Sample	30	5.547	1.341	1.3	4.7	5.6	6.6	8.2
2: SRS Sample	27	4.356	0.556	3.3	3.9	4.3	4.7	5.4

Biased sample overestimate the population mean. Increasing sample size will not reduce the bias.

SRS sample estimate the population mean and SRS samples means will be centered around the population mean

Using Random Digit Table to get Random Numbers to Draw Your Sample

You start on any line

You choose your numbers according to the maximum digits that is used to the population size

Ex: if your population size is 695, then you go 3 digit by 3 digit

You discard any repeated number or any number larger than the population size

If the line ends you continue on the next line

Ex: Choose a sample of 30 from the 695 population

I will use line 3:

345, 689, 564, 153, 077, 113, 315, 441, 424, 429, 319, 195, 305, 691, 116, 227, 014, 173, 465, 680,
207, 333, 627, 187, 040, 565, 258, 215, 529, 307

Line										
1	17139	27838	19139	82031	46143	93922	32001	05378	42457	94248
2	20875	29387	32582	86235	35805	66529	00886	25875	40156	92636
3	34558	95648	79767	15307	71133	15714	44142	44293	19195	30569
4	11169	41277	01417	34656	80207	33362	71878	31767	04056	52582
5	15529	30766	70264	86253	07179	24757	57502	51033	16551	66731
6	33241	87844	41420	10084	55529	68560	50069	50652	76104	42086
7	83594	48720	96632	39724	50318	91370	68016	06222	26806	86726
8	01727	52832	80950	27135	14110	92292	17049	60257	01638	04460
9	86595	21694	79570	74409	95087	75424	57042	27349	16229	06930
10	65723	85441	37191	75134	12845	67868	51500	97761	35448	56096

Some useful definitions to remember and habits to develop from this topic are

- Take care when selecting a sample from a population if you hope to generalize results from the sample to describe the population.
- Sampling bias occurs when a sampling method favors certain outcomes over others. Bias is a property of the sample selection method, not simply of a single sample.
- **Simple random sampling** eliminates sampling bias by giving every observational unit in the population an equal chance of being selected. You discovered this property by hypothetically taking multiple simple random samples and seeing that the sample statistics *center* around the population parameter rather than systematically tending to over- or underestimate the population parameter.
- **Sampling variability** means that different samples produce different results (i.e., different values of sample statistics). Even with simple random samples, you expect results to vary from sample to sample, but with a predictable long-term pattern.
- Increasing the sample size reduces sampling variability. In other words, larger samples produce more **precision**, less variation in results from sample to sample. But if the sampling method is biased in the first place, then taking a larger sample does not reduce bias.
- The size of the *population* is not relevant to the issues of sampling bias or precision (as long as it is much larger than the size of the sample).

You should be able to

- **Explain what sampling bias means, and anticipate the direction of sampling bias that might arise based on a particular sampling method. (Activity 4-1)**
- **Explain what simple random sampling means and why it eliminates sampling bias. (Activity 4-2)**
- **Implement random sampling to select a sample from a population. (Activity 4-2)**
- **Describe properties of random sampling, including sampling variability, unbiasedness, greater precision from larger random samples, and the irrelevance of population size on precision. (Activities 4-3, 4-4)**

Make sure you understand the difference between precise and unbiased.

A random sample is necessary to get an unbiased statistic. Statistics from different random samples from the same population are unbiased if they are centered at the population parameter. We expect different sample means taken randomly from the same population to be centered at the population mean. Sample statistics from larger samples are more precise and closer together.

If the sample is large and random, the sample statistic is a more accurate estimate of the corresponding population parameter. We expect a sample mean calculated from a large, random sample to be precise and unbiased estimate of the population mean.

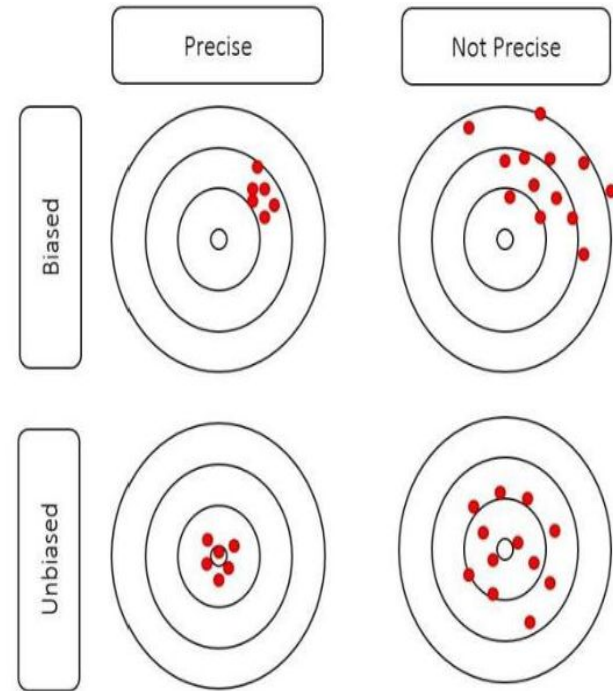


Figure 2.1: Precision & Bias

Topic 4: Random Sampling

- **Random Sample**

- Every member of the population has the same chance of being selected for a sample

- **Simple Random Sample**

- Every possible sample (of sample size n) has the same chance of being selected

- **Example:**

- If we pick 30 TJ students for a study.
- Random sampling is to use all TJ students and randomly pick 30
- Simple Random sampling is to identify all samples of size 30 and then randomly pick one of those samples

Topic 4: Random Sampling

- **Sampling Variability**

- Each sample produces a different sample statistic
- Example:
 - Pick a sample of 30 TJ students and calculate mean weight ($\bar{x}_1$)
 - Pick another sample of 30 TJ students and calculate mean weight ($\bar{x}_2$)
 - Do you think $\bar{x}_1 = \bar{x}_2$?

- **Precision**

- How much the statistic varies from sample to sample
 - How different are $\bar{x}_1$ and $\bar{x}_2$?
- Related to sample size

Topics 9 Review

Standard Deviation: is how much a typical data value in a distribution differs from the mean of the sample. It is calculated by finding the average of the square distances and then taking the square root. Variance is standard deviation, squared.

Interpretation of Standard Deviation: Context, A typical “observational unit” “variable name” is “SD value and measuring units below or above the mean of “variable name and observational units which is “Mean Value and units of measurements”

In Dr. Ebrahim class period 3, a typical student’s height is 2 inches below or above the mean height of the students in the class which is 70 inches

Standard deviation does not depend on the range only it is more affected by how the data points distributed over the range.

For the same range the distribution that has more data points near the minimum and the maximum values will have higher standard deviation than the distribution that has more data points at the center.

Measure of Center:

Mean, Median or Mode (Mode can be also used for categorical data)

Measure of Spread:

Range, IQR or Standard Deviation (has to be expressed as single numbers not intervals)

Range = Max - Min

$IQR = Q_3 - Q_1$

Standard deviation is how far a typical observational unit value is above or below the mean.

Standard deviation value does not depend on the range only it depends more on how many points near the center and how many point away from the center.

Median and IQR are resistant measures which means they are not affected by outliers.

Mean, SD and range are affected by the outliers.

Identify different shapes of these data sets and find which data set has smaller standard deviation. Use the study guide to get the answer

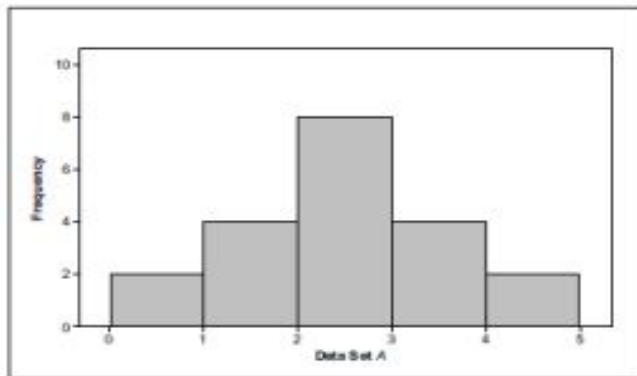


Figure 6.5. Histogram of Data Set A.

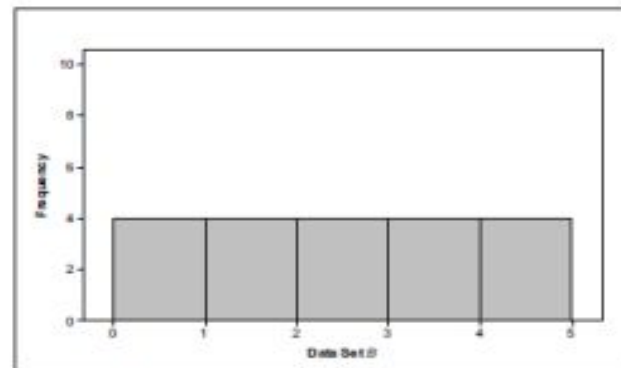


Figure 6.6. Histogram of Data Set B.

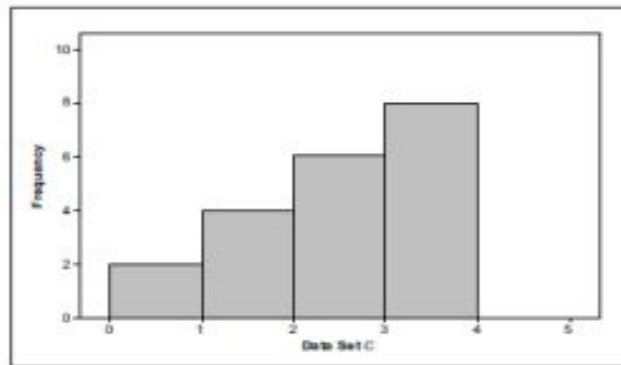


Figure 6.7. Histogram of Data Set C.

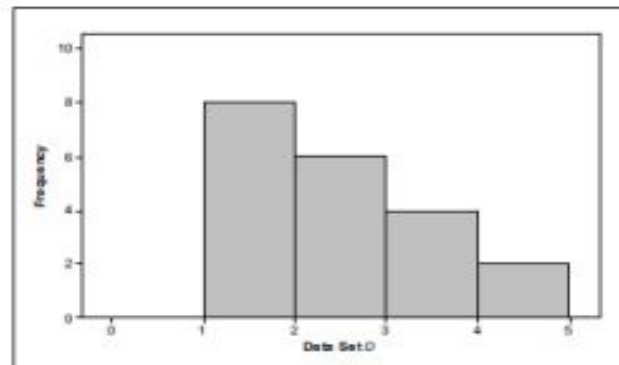


Figure 6.8. Histogram of Data Set D.

Identify different shapes of these data sets and find which data set has smaller standard deviation.
Use the study guide to get the answer

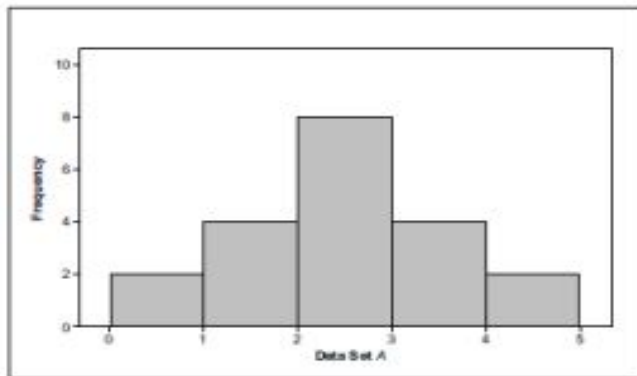


Figure 6.5. Histogram of Data Set A.

$$A < B$$

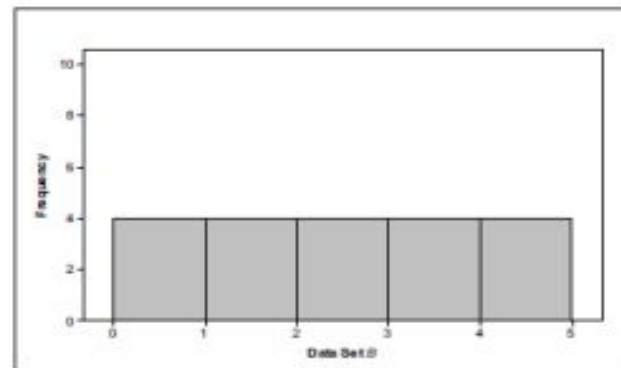


Figure 6.6. Histogram of Data Set B.

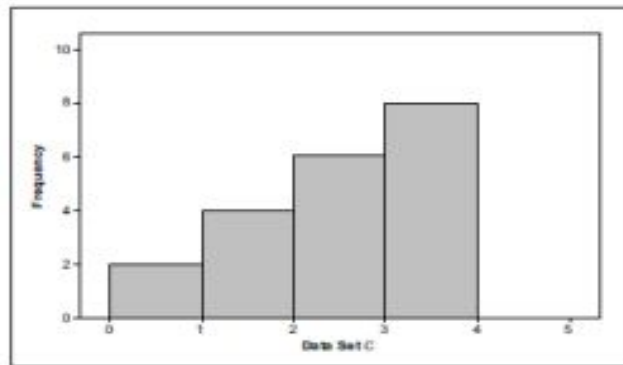


Figure 6.7. Histogram of Data Set C.

$$C = D$$

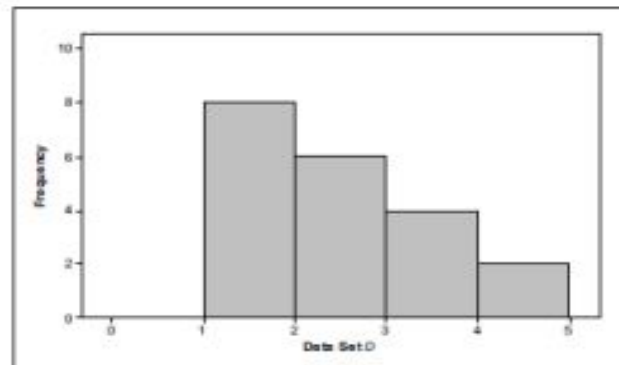
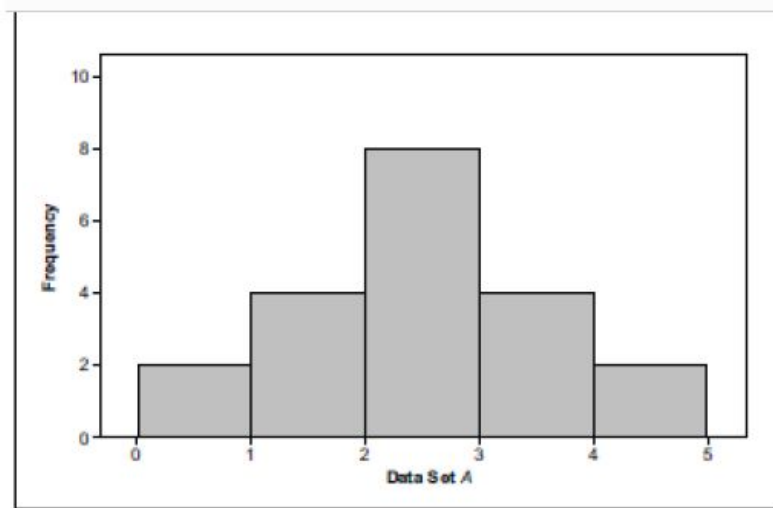
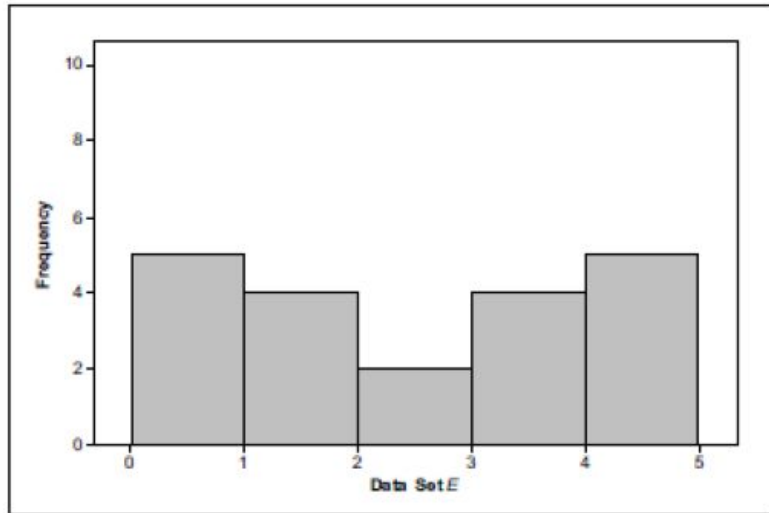
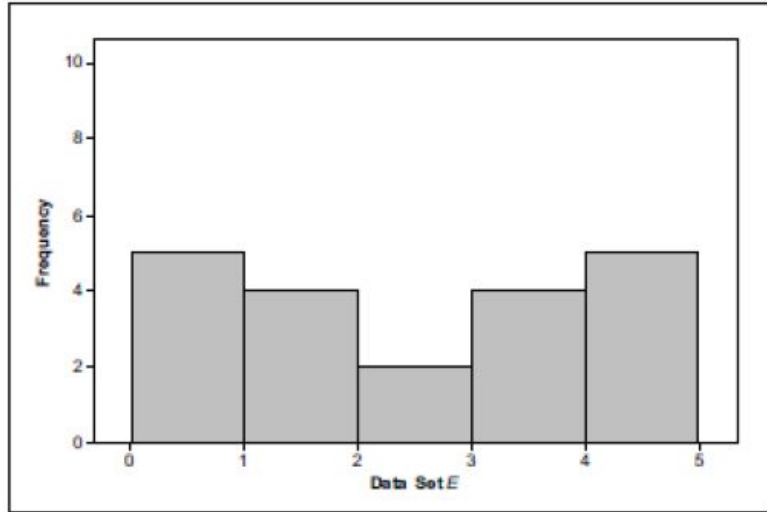


Figure 6.8. Histogram of Data Set D.

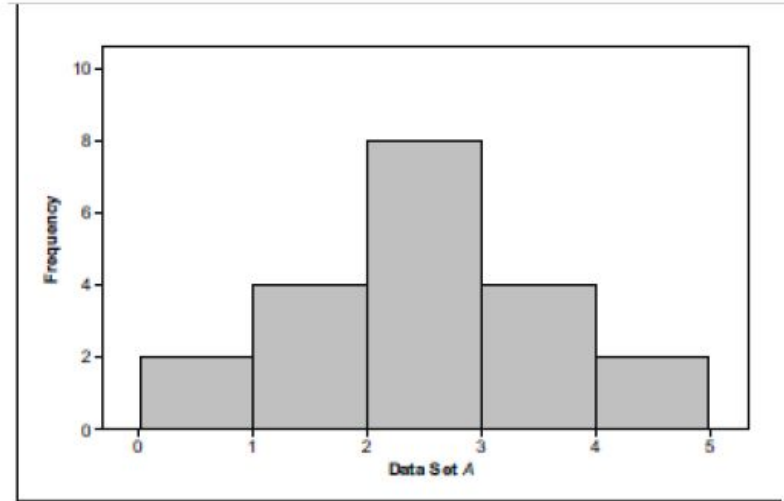
Identify different shapes of these data sets and find which data set has smaller standard deviation.
Use the study guide to get the answer



Identify different shapes of these data sets and find which data set has smaller standard deviation.
Use the study guide to get the answer



$$E > A$$



When moving data point from the center of the distribution to the terminal sides the standard deviation will increase

(More data away from the center means higher standard deviation)

- Variability does not measure either bumpiness nor variety/distinct values; variability measures spread from the center (mean). A distribution can be very 'bumpy' without having a great deal of variability and vice versa. It is more important to consider the overall tendency for data values to be near or far from the center in determining variability.
- Adding or subtracting certain value to/from each data value will increase/decrease the median and the mean with the same value
- But standard deviation and IQR will not change
- If you multiply each data value by certain factor then new mean, median, SD and IQR will be multiplied by the same factor

Complete Activity 9-3 and **Exercise 9-13 in your Google Doc.**

It is about:

- Adding or subtracting certain value to/from each data value will increase/decrease the median and the mean with the same value
- But standard deviation and IQR will not change
- If you multiply each data value by certain factor then new mean, median, SD and IQR will be multiplied by the same factor

Watch Out!

Turn to page 178 and answer the following questions based on the Watch Out

1. If there is an odd number of observations, is the median included in the upper half of the data, the lower half or neither?

Do not include the median in either the upper or lower half.

2. Can a range or interquartile range be reported as an interval? **No.**

3. Can range and interquartile range be used with categorical data? **No.**

If each data value is increased by certain value A
The new mean of the data set will be (Old mean + A),
but the standard deviation will stay the same

You have to be able estimate the median if a frequency table is given

Median
located in this
interval

Height (inch)	50-53	54-57	58-61	62-65	66-69	>69
Frequency	22	31	11	8	1	2

Topics 1 & 2 Review

Data: the numbers or categories recorded for the observational units in a study

Observational Unit: the person or thing to which the variable number or category is assigned such as a student in your class

Variable: any characteristic of a person or thing that can be assigned a number or category

Quantitative Variable: a variable that measures a numerical characteristic such as height

Categorical Variable: a variable that records a group designation such as gender

Binary Categorical Variable: a categorical variable with only two possible categories, for example, left or right

Variability: the phenomenon of a variable taking on different values or categories from one observation unit to another

Research Question: a question that looks for patterns in a variable or compares a variable across different groups or looks for a relationship between variables

Distribution: the pattern of variation of a variable; with a categorical variable, distribution means the variable's possible categories and the proportion of responses in each

Dotplot: a display of the distribution of relatively small data sets where each data point is represented by a dot.

Bar Graph: a graph that displays the distribution of a categorical variable

Consistency: this refers to how variable, or spread out, the values in a dataset are for a quantitative variable

Tendency vs. Consistency

Now read page 22 in the textbook.

1. Define tendency and consistency in your own words.

Tendency refers to the center of the distribution; consistency (or variability) to the spread in the distribution. You should always look for both.

2. State the difference, in your own words, between tendency and consistency.

Tendency is what a typical observational unit is and consistency is how likely a typical observational unit will be that.

Important Points To Remember

Tendency refers to the center of a distribution

Both variability and consistency can be used when talking about the spread but they have two different meanings, high variability means large spread while high consistency means small spread.

When labelling a graph, indicate the variable on the horizontal axis.

When labelling a graph, indicate units such as count, proportion, or percentage on the vertical axis.

It is necessary to include units of measurements when labelling axes.

An example of quantitative data is your age on the first day at TJ.

An example of categorical data is the type of car your parents drive.

The most fundamental concept of statistics is variability.

In a dotplot, a dot represents data from one observational unit.

Dotplots pertain to a quantitative variable; bar graphs to a categorical variable.

DATA TYPES:

Quantitative

- Can you Measure it?
- Can you count it?

Visualize w/ Dot Plot

Look how a good graph has to look like

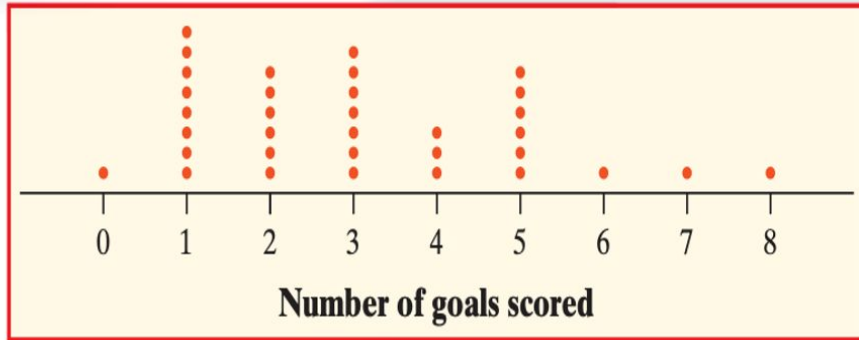
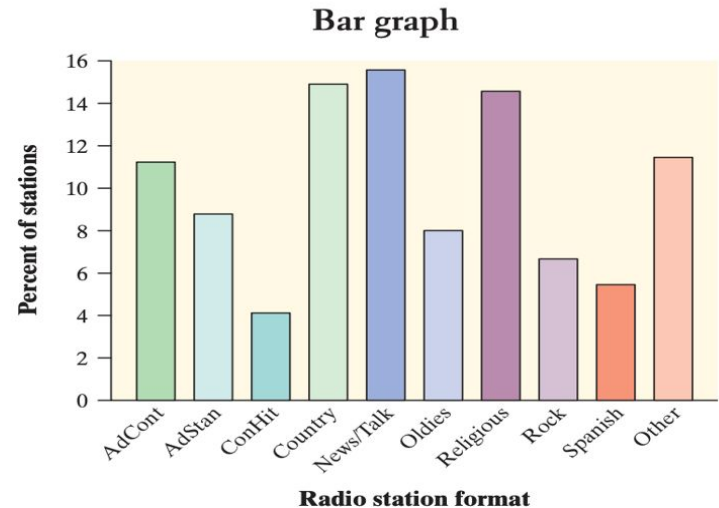


FIGURE 1.9 A dotplot of goals scored by the U.S. women's soccer team in 2004.

Categorical / Binary

Multiple Choices

- Types of Something
 - Binary (2): Yes or No
- Visualize w/ Bar Graph



DATA TYPES:

Quantitative

- Can you Measure it?
- Can you count it?

Visualize w/ Dot Plot

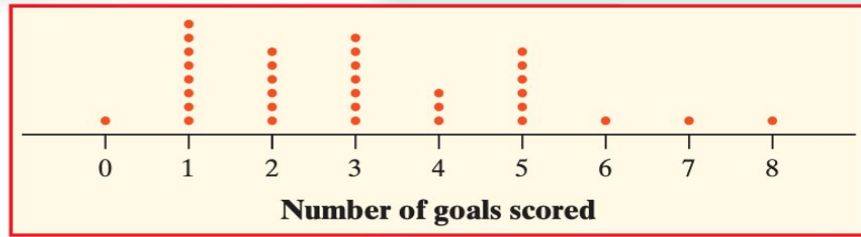


FIGURE 1.9 A dotplot of goals scored by the U.S. women's soccer team in 2004.

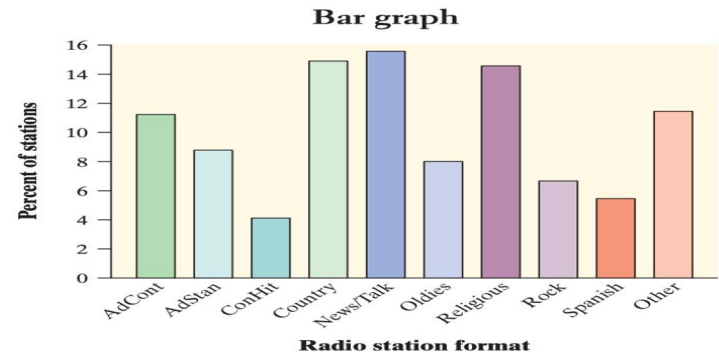
For quantitative data, horizontal axis supposed to have:

- A scale with numbers
- Label (name of the variable)
- Units of measurements

Categorical / Binary

- Multiple Choices
- Types of Something
- Binary (2): Yes or No

Visualize w/ Bar Graph



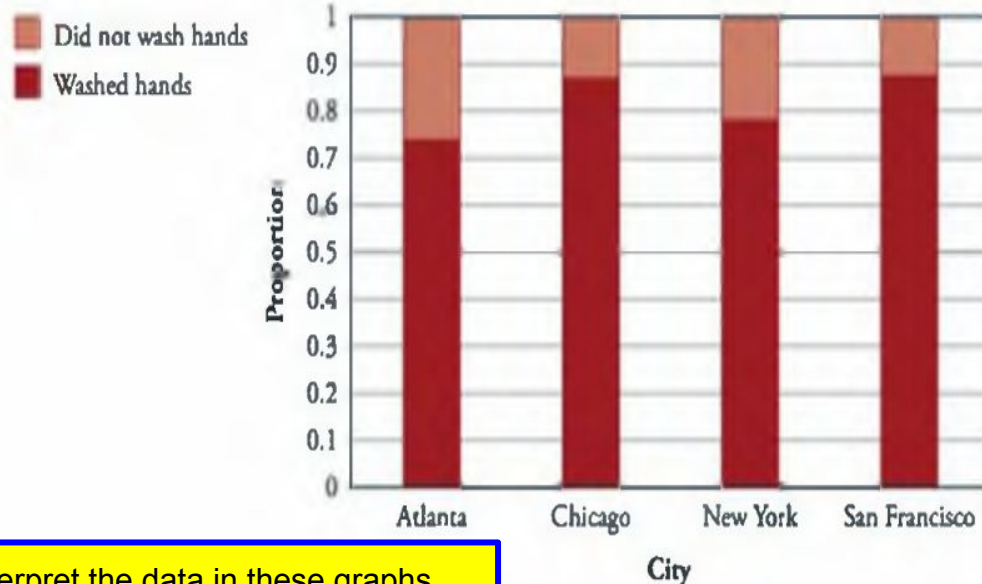
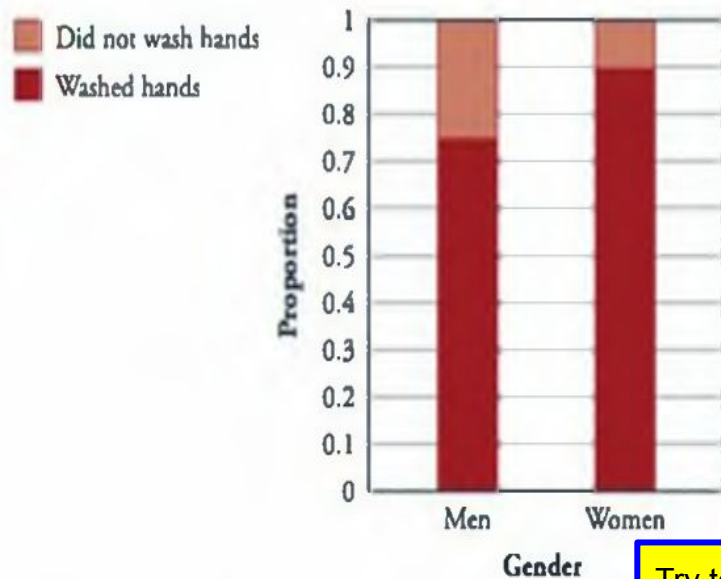
For categorical data, horizontal axis supposed to have:

- Categories
- Label (name of the variable)

Each bar has a different color

Vertical axis has the percentage, frequency or count

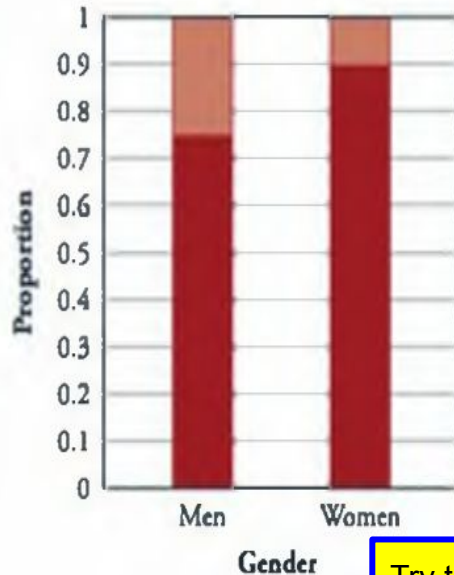
The visual display that you have constructed is called a **bar graph**. Bar graphs display the distribution of a categorical variable. The **distribution** of a variable refers to its pattern of variation. With a categorical variable, distribution means the variable's possible categories and the proportion of responses in each.



Try to Interpret the data in these graphs

The visual display that you have constructed is called a **bar graph**. Bar graphs display the distribution of a categorical variable. The **distribution** of a variable refers to its pattern of variation. With a categorical variable, distribution means the variable's possible categories and the proportion of responses in each.

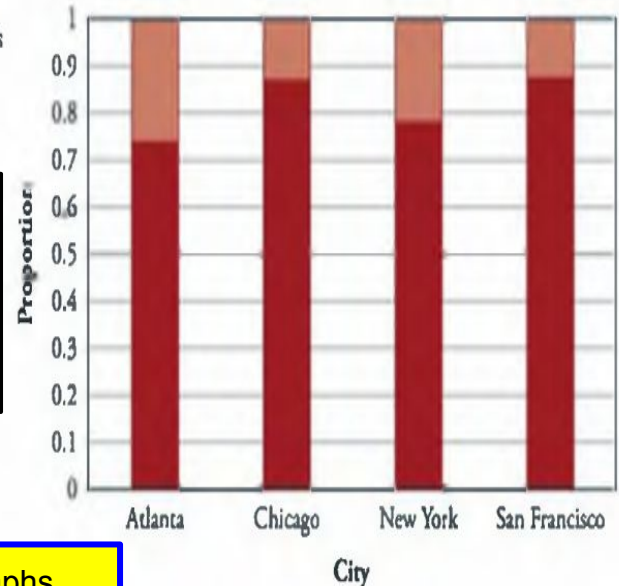
Did not wash hands
Washed hands



0.75 of men
washed hands

0.25 of men did not
wash hands

Did not wash hands
Washed hands



0.85 of Chicago
washed hands

0.15 of Chicago did not
wash hands

Try to Interpret the data in these graphs

The visual display that you constructed for states visited is a **dotplot**. Dotplots are useful for displaying the distribution of relatively small datasets of a quantitative variable.



Watch Out

- Each dot in a dotplot represents a different observational unit.
- Develop the habit of always labeling the axis with the name of the variable.
- Dotplots can be cumbersome to create by hand with larger datasets, so you can often use technology to create the graph.
- Dotplots pertain to a quantitative variable; bar graphs, to a categorical variable.

The visual display that you constructed for states visited is a **dotplot**. Dotplots are useful for displaying the distribution of relatively small datasets of a quantitative variable.

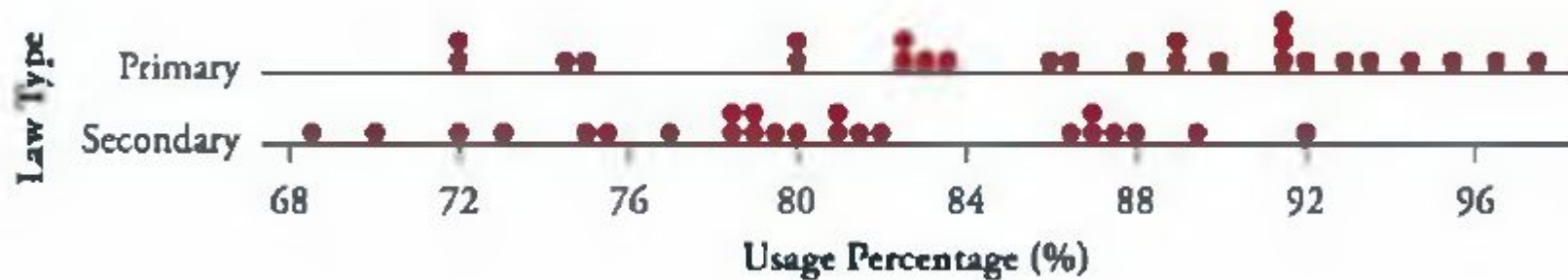


A backpack weight tends to be 11 pounds and there is high variability

- Each dot in a dotplot represents a different observational unit.
- Develop the habit of always labeling the axis with the name of the variable.
- Dotplots can be cumbersome to create by hand with larger datasets, so you can often use technology to create the graph.
- Dotplots pertain to a quantitative variable; bar graphs, to a categorical variable.

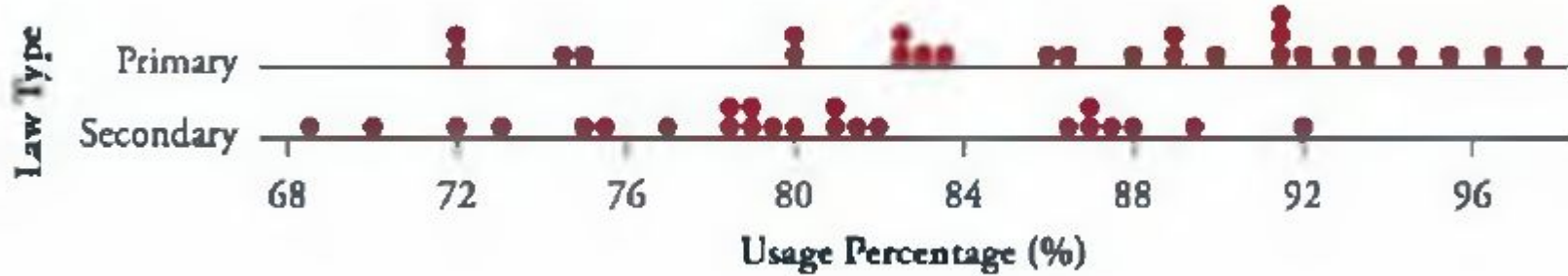
1- write a research question.

2- What can be said about the top usage percentage?



1- write a research question.

2- what can be said about the top usage percentage



1- Research question:

Do the primary states have higher usage percentage than secondary states?

2- The top usage percentage of 98% happened in a primary state

Try to Interpret the data in these distributions

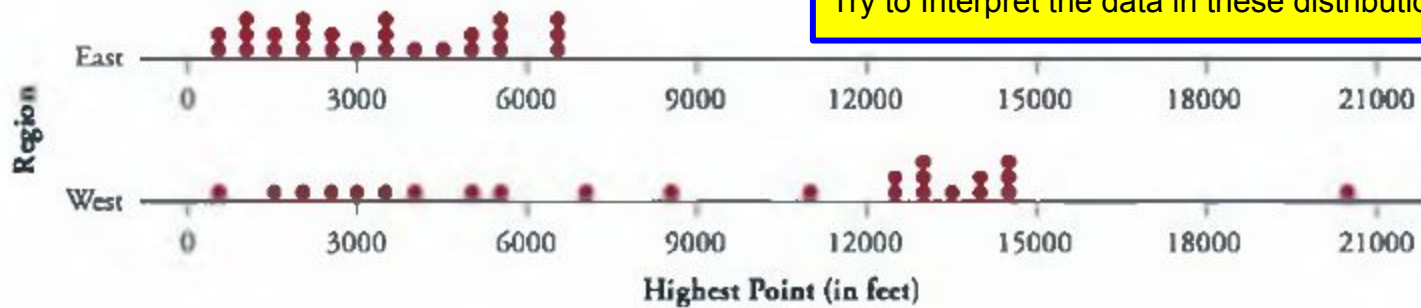


Figure Ex. 2-15

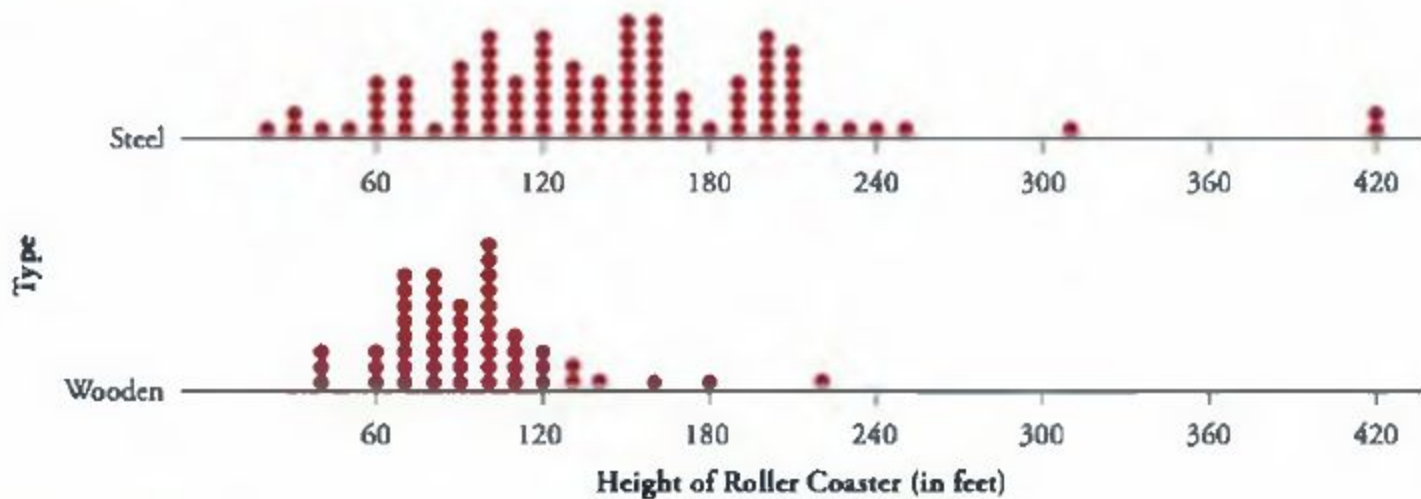


Figure Ex. 2-17

Try to Interpret the data in these distributions



Figure Ex. 2-15

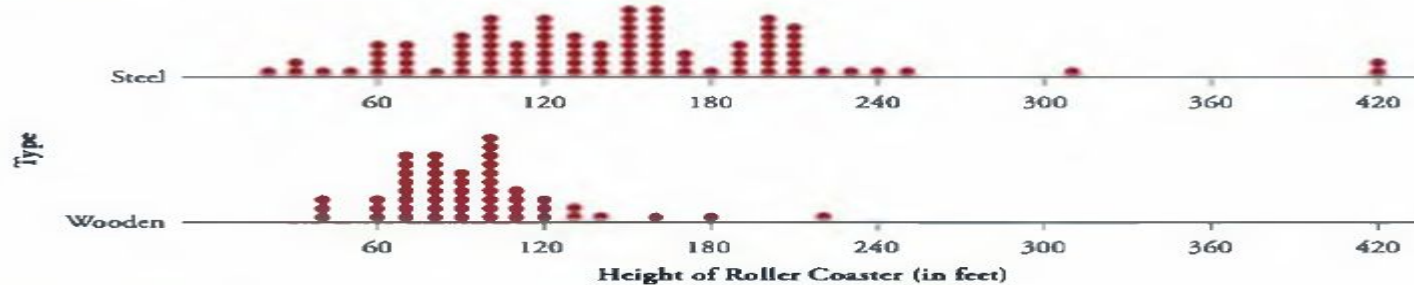


Figure Ex. 2-17

The West region tends to have bigger highest point roller coasters than the East region and West region has more variability than East region.

The steel roller coasters tends to have more height than the wooden ones but the wooden roller coasters have more consistency than the steel ones.

Exercise 2-17: Roller Coasters 2-17, 10-2

The Roller Coaster DataBase (www.rcdb.com) contains extensive data on roller coasters around the world. Some of the variables for which data are provided include height, length, speed, number of inversions, type (steel or wooden), and design (sit-down, stand-up, inverted).

- a. What are the observational units for these data?
- b. Classify each of the variables listed in the previous paragraph as categorical (also binary?) or quantitative.

Exercise 2-17: Roller Coasters 2-17, 10-2

The Roller Coaster DataBase (www.rcdb.com) contains extensive data on roller coasters around the world. Some of the variables for which data are provided include height, length, speed, number of inversions, type (steel or wooden), and design (sit-down, stand-up, inverted).

- What are the observational units for these data?
- Classify each of the variables listed in the previous paragraph as categorical (also binary?) or quantitative.

Watch Out!

Read the Watch Out on page 18 and answer below:

1. What does each dot in a dot plot represent? **A different observational unit**
2. What type of variable is displayed in dot plots? **Quantitative**
What type of variable is displayed in bar graphs? **Categorical**

Now read page 21 in the textbook. When are dotplots and bar graphs most illuminating? **When used to compare the distribution of a variable between two or more groups.**

Tendency vs. Consistency

Now read page 22 in the textbook.

1. Define tendency and consistency in your own words.
Tendency refers to the center of the distribution; consistency (or variability) to the spread in the distribution. You should always look for both.
2. State the difference, in your own words, between tendency and consistency.
Tendency is what a typical observational unit is and consistency is how likely a typical observational unit will be that.

Topics 3 Review

Observational Units

Quantitative Variables

Categorical Variables

Explanatory (Independent) Variable

Response (dependent) Variable

Experiment

Observational Study

Cause-and-Effect Relationship

Association

Conclusion

Population (Parameter)

Sample (Statistics)

Sample Size

Random Sample

Sampling Bias

Convenience Sampling

Voluntary Response Sampling

Nonresponse Bias

Response Bias

Lurking Variables

Confounding Variables

Simple Random Sampling (SRS)

The **population** in a study refers to the *entire* group of people or objects (observational units) of interest. A **sample** is a (typically small) *part* of the population from whom or which data are gathered to learn about the population as a whole. If the sample is selected carefully, so that it is **representative** of (has similar characteristics to) the population, you can learn useful information. The number of observational units (people or objects) studied in a sample is the **sample size**.

If I want to study the population of the freshmen I can take a sample of the students who are taking classes with me. But this could be biased sample.

In selecting a sample from a population, ideally we would start with a **sampling frame**, an actual list of every member of the population that we want to sample from.

A sampling procedure is said to have **sampling bias** if it tends systematically to overrepresent certain segments of the population and to underrepresent others.

A **parameter** is a number that describes a population, whereas a **statistic** is a number that describes a sample.

The variable whose effect you want to study is the **explanatory variable**. The variable that you suspect is affected by the other variable is the **response variable**. The response variable is often considered the outcome of interest, but we think those outcomes may be influenced by the explanatory variable. Often the explanatory variable comes first in a time sequence.

Variables that are not considered in the study but that may also be related to the response variable are called **lurking variables**. Some of these lurking variables may also differ among the explanatory variable groups in such a way that we cannot distinguish their effects from those of the explanatory variable. These **confounding variables** prevent us from drawing a cause-and-effect conclusion between the explanatory and response variables.

In an **observational study**, researchers **passively** observe and record information about observational units. An observational study does not control for the possible effects of confounding variables, and therefore, although an observational study may establish an association or relationship between the explanatory and response variables, **you cannot establish a cause-and-effect conclusion.**

An **experiment** is a study in which the experimenter *actively* **imposes the treatment** (explanatory variable group) on the subjects. Ideally, the groups of subjects are identical in all respects other than the explanatory variable so that the researcher can then see the explanatory variable's direct effects on the response variable.

Observational Study and Experiment.

In Observational study: You do not impose any treatment. You are just observing the observational units **passively** or asking questions (survey). You end up with association. You can not avoid the effect of the confounding variables. Most of the time people choose to participate which will make the association doubtful.

You end up with **association conclusion**

In Experiment: You **impose a treatment**. You control what is happening. You end up with **cause and effect conclusion**

So, to be able to draw a cause-and-effect conclusion you must do experiment and you must have two groups (control group and treatment group)

To generalize your finding you need SRS (Representative Sample)

To draw a cause and effect conclusion, you need experiment (Random Assignment)

Whatever you are doing observational study or experiment

- 1- your sample has to be representative of the population this can be done by simple random sampling (usually it is hard to do it for experiment) (Topic 4)
- 2- The explanatory variable is the treatment. It is the independent variable
- 3- The response variable is the outcome. It is the dependent variable
- 4- You have to eliminate the effect of the confounding variables using Random Assignment. (Topic 5)
- 5- Confounding variables connected to the explanatory variable and to the response variable
- 6- You have to differentiate between random sampling and random assignment: They are different and has different objectives

Some useful definitions to remember and habits to develop from this topic are

- The **population** is the entire collection of observational units (people or objects) of interest; the **sample** is the part of the population on which you gather data.
- A **parameter** is a number that describes a population; a **statistic** is a number that describes a sample. Notice that these are numbers, not people or objects as population and sample are.
- The **sampling frame** is a list of all observational units in the population, ideally used for selecting a sample from the population.
- An **explanatory variable** is one whose effect you study; a **response variable** is the outcome you record.
- **Sampling bias** is the systematic tendency of a sampling method to overrepresent some parts of the population and underrepresent others. This prevents us from **generalizing** from the sample to the population.
- An **observational study** is one in which researchers record information passively, without attempting to impose the explanatory variable on the observational units.
- A **confounding variable** is one whose effect on the response variable cannot be distinguished from the effect of the explanatory variable. To properly identify a confounding variable, discuss how it is related to both the explanatory and the response variables.
- You cannot draw **cause-and-effect conclusions** from observational studies because other factors (confounding variables) might differ between the groups.

You should be able to

- Identify the population and sample, parameter, and statistic in a study. (Activity 3-1)
- Recognize and identify sampling bias in a study. (Activity 3-1)
- Think about the appropriate population to which sample results can be generalized, based on how the sample was selected. (Activity 3-2)
- Identify a potentially confounding variable in an observational study, and explain how the issue of confounding prevents drawing cause-and-effect conclusions from observational studies. (Activities 3-3, 3-4)
- Classify variables in a study as explanatory or response variables. (Activity 3-4)

Example 1

At a certain university, students who live in the dormitories eat at a common dining hall. Recently, some students have been complaining about the quality of the food served there. The dining hall manager decided to do a survey to estimate the proportion of students living in the dormitories who think that the quality of the food should be improved. One evening, the manager asked the first 100 students entering the dining hall to answer the following question:

Many students believe that the food served in the dining hall needs improvement. Do you think that the quality of the food served here needs improvement, even though that would increase the cost of the meal plan?

_____ Yes

_____ No

_____ No opinion

- a. In this setting, explain how bias may have been introduced based on the way this sample was selected **and** suggest how the sample could have been selected differently to avoid that bias.

- b. In this setting, explain how bias may have been introduced based on the way the question was worded **and** suggest how it could have been worded differently to avoid that bias.

Example 1

At a certain university, students who live in the dormitories eat at a common dining hall. Recently, some students have been complaining about the quality of the food served there. The dining hall manager decided to do a survey to estimate the proportion of students living in the dormitories who think that the quality of the food should be improved. One evening, the manager asked the first 100 students entering the dining hall to answer the following question:

Many students believe that the food served in the dining hall needs improvement. Do you think that the quality of the food served here needs improvement, even though that would increase the cost of the meal plan?

_____ Yes

_____ No

_____ No opinion

a. In this setting, explain how bias may have been introduced based on the way this sample was selected **and** suggest how the sample could have been selected differently to avoid that bias.

Since the manager used a convenience sample (the first 100 students entering the cafeteria), bias may have been introduced. Students who arrive at the cafeteria early may have opinions of food quality that differ in some important way from other students who live in the dormitories. This bias could be avoided by selecting a random sample of 100 dormitory residents instead of just asking the first 100 students entering the cafeteria.

b. In this setting, explain how bias may have been introduced based on the way the question was worded **and** suggest how it could have been worded differently to avoid that bias.

The way the question is worded may be leading. The first part of the question included a statement that many students think the food needs improvement. This may lead people to support this view by responding that the food does need improvement. The inclusion of the phrase "even though that would increase the cost of the meal plan" may lead students to say the food is OK only because they do not want to pay more. A better wording might be to simply ask "Do you think that the quality of the food served in the cafeteria needs improvement?"

Example 2

A florist wanted to investigate whether a new powder added to the water of cut flowers helps to keep the flowers fresh longer than just water alone. For a shipment of roses that was delivered to the store, the florist flipped a coin before placing each rose in its own individual container with water. If the coin landed heads up, the rose was placed in water with the new powder; otherwise, the rose was placed in water alone.

- a) Is this an observational study or an experiment? Why
- b) What are the observational units in this study?
- c) What is the treatment?
- d) What is the explanatory variable?
- e) What is the response variable?

Example 2

A florist wanted to investigate whether a new powder added to the water of cut flowers helps to keep the flowers fresh longer than just water alone. For a shipment of roses that was delivered to the store, the florist flipped a coin before placing each rose in its own individual container with water. If the coin landed heads up, the rose was placed in water with the new powder; otherwise, the rose was placed in water alone.

- a) Is this an observational study or an experiment? Why
This is an experiment. The florist is imposing a treatment.
- b) What are the observational units in this study?
The observational units are the roses.
- c) What is the treatment?
The treatment is the new powder added to the water
- d) What is the explanatory variable?
The explanatory variable is the treatment (water alone / water with new powder), Categorical, Binary
- e) What is the response variable?
The response variable is time the roses stay fresh

Example 3

A study will be conducted to determine whether children learn their multiplication facts better by practicing with flash cards or by practicing on a computer. Children who volunteer for the study will be randomly assigned to one of the two treatments. After practice, the children will be given a test on their multiplication facts.

- a) Is this an observational study or an experiment? Why
- b) What are the observational units in this study?
- c) What was the treatment?
- d) What was the explanatory variable?
- e) What was the response variable?

Example 3

A study will be conducted to determine whether children learn their multiplication facts better by practicing with flash cards or by practicing on a computer. Children who volunteer for the study will be randomly assigned to one of the two treatments. After practice, the children will be given a test on their multiplication facts.

- a) Is this an observational study or an experiment? Why
- b) What are the observational units in this study?
- c) What was the treatment?
- d) What was the explanatory variable?
- e) What was the response variable?
- f) Was double blindness involved in this study?

Solve this example by yourself

1. Population vs. Sample

- **Population:** The **entire group** of individuals or items that you want to draw conclusions about.
 - **Example:** *All teenagers living in a specific school district or every single registered voter in the country.*
- **Sample:** A **smaller subset** chosen from the population to represent the whole group.
 - **Example:** *50 students randomly selected from one high school in that district.*

2. Parameter vs. Statistic

- **Parameter:** A numerical summary value that describes a **characteristic of the entire population**. It is usually fixed, but often unknown because measuring an entire population is difficult or impossible.
 - **Example:** *The true average height of ALL teenagers in the country (μ) or the exact percentage of ALL voters who support a candidate (p).*
- **Statistic:** A numerical summary value calculated from a **sample**. It is known once measured, but can vary depending on which sample you pick. We use statistics to estimate parameters.
 - **Example:** *The average height of the 50 sampled students ($\bar{x}$) or the percentage of voters in a 1,000-person poll who support a candidate ($\hat{p}$).*

1. Observational Study vs. Experiment

- **Observational Study:**
 - **What it is:** Researchers observe and record data **without imposing any treatment or intervening** in the subjects' lives.
 - **Key Limitation:** **Cannot prove causation** due to potential confounding variables.
 - **Example:** Tracking people who naturally choose to drink green tea vs. those who don't, then measuring their cholesterol levels over 5 years.
- **Experiment:**
 - **What it is:** Researchers **deliberately impose a treatment** (manipulate variables) and use **random assignment** to divide subjects into groups (e.g., treatment group vs. control group receiving a placebo).
 - **Key Advantage:** **Can establish cause-and-effect relationships** because random assignment balances out confounding variables.
 - **Example:** Randomly assigning 100 participants to take either a real green tea extract pill or a sugar placebo pill daily for 8 weeks to measure cholesterol changes.

2. Explanatory Variable vs. Response Variable

- **Explanatory Variable (Independent Variable / X):**
 - The variable that is manipulated (in an experiment) or categorized (in an observational study). It is hypothesized to explain or predict the outcome.
 - **Example:** *Daily dose of green tea extract (e.g., 0 mg vs. 500 mg).*
- **Response Variable (Dependent Variable / Y):**
 - The variable that is measured as the outcome or result of the study.
 - **Example:** *The change in blood cholesterol levels at the end of the study.*

Summary Checklist

Concept

Observational Study

Experiment

Treatment Imposed?

 No (Just observing)

 Yes (Deliberately applied)

Random Assignment?

 No

 Yes

Can Prove Causation?

 No (Only association/correlation)

 Yes (Cause-and-effect)

EXAMPLE

Parameter versus Statistic

Suppose the percentage of all students on your campus who have a job is 84.9%. This value represents a parameter because it is a numerical summary of a population.

Suppose a sample of 250 students is obtained, and from this sample we find that 86.4% have a job. This value represents a statistic because it is a numerical summary based on a sample.

EXAMPLE *The Characteristics of an Experiment*

The English Department of a community college is considering adopting an online version of the freshman English course. To compare the new online course to the traditional course, an English Department faculty member randomly splits a section of her course. Half of the students receive the traditional course and the other half is given an online version. At the end of the semester, both groups will be given a test to determine which performed better.

- (a) Who are the experimental units? **The students in the class**
- (b) What is the population for which this study applies? **All students who enroll in the class**
- (c) What are the treatments? **Traditional vs. online instruction**
- (d) What is the response variable? **Exam score**
- (e) Why can't this experiment be conducted with blinding?
 Both the students and instructor know which treatment they are receiving

Both lurking and confounding variables distort statistical results, but they differ in where they hide and how they interfere with your ability to prove cause and effect.

Feature	Lurking Variable	Confounding Variable
Study Presence	Unmeasured / Hidden outside the study	Measured or present alongside X
Primary Effect	Creates a false (spurious) link between X and Y	Tangled up with X , so you can't isolate the real cause
Mechanism	Influences both X and Y secretly	Varies along with X and directly impacts Y

Lurking Variables: The Hidden Driver

- **What it is:** A variable that was completely omitted or unmeasured in a study, yet it secretly influences both the explanatory variable (X) and the response variable (Y).
- **Effect:** It makes X and Y appear related when there is no direct causal link between them.
- **Classic Example:** A study finds a strong positive correlation between ice cream sales (X) and shark attacks (Y). The lurking variable is **outdoor temperature**—hot summer weather causes people to buy ice cream and go swimming at the same time, but ice cream doesn't attract sharks.

Confounding Variables: The Tangled Interferer

- **What it is:** A variable that is closely tied to your explanatory variable (X) and also impacts your response variable (Y). Because its effect overlaps with X , you cannot separate how much of the outcome came from X versus the extra variable.
- **Effect:** The effects of X and the confounding variable are "confounded" (mixed together).
- **Classic Example:** An experiment tests if a new tutoring program (X) improves math test scores (Y). If the tutoring group consists entirely of honors students while the control group consists of struggling students, **prior student ability** is a confounding variable. You can't tell whether high test scores came from the tutoring or the students' existing intelligence.

Topics 7, 8 & 10 Review

Mean: the arithmetic average or balance point of a distribution.

Median: the middle value in a distribution; often considered the “typical” value.

Mode: the most commonly occurring value in a distribution.

Mound shaped distribution: single peak is at the center of the distribution

Symmetric distribution: left side of a distribution is roughly a mirror image of the right side

Skewed left: the tail of the distribution follows the smaller values towards the left

Skewed right: the tail of the distribution follows the larger values towards the right

Consistency: this refers to how variable, or spread out, the values in a dataset are for a quantitative variable

Resistant: When a measure's value is relatively unaffected by the presence of outliers.

Interquartile range: the difference between the upper quartile and lower quartile.

Boxplot: the graph that illustrates the five-number summary. The boxes are drawn between the quartiles and median. The whiskers extend from the quartiles to the min or max.

Lower quartile: the 25th percentile or the value such that 25% of the observations fall below it and 75% fall above it.

Upper quartile: the 75th percentile or the value such that 75% of the observations fall below it and 25% fall above it.

Five-number summary: the minimum value, lower quartile, median, upper quartile, and maximum value.

Modified boxplot: a specialized boxplot that conveys additional information by treating outliers differently. On these graphs, you mark outliers using a special symbol and then extend the boxplot's whiskers only to the most extreme non-outlier value. (This is the boxplot we will use this semester; we won't refer to it as modified.)

Split stemplot: This type of stemplot is used when too few stems and important details can be lost because the data are clumped together. A split stem plot that displays each stem twice, where the 0-4 leaves appear on the first stem and the 5-9 leaves appear on the second

Side-by-side stemplot: a stemplot that is used to compare two sets of data where a common set of stems is placed in the middle of the display with leaves out in either direction

Histogram: a graphical display similar to a dotplot or stemplot, but more feasible when displaying very large data plots. Bars are constructed whose heights correspond to the frequency in each subinterval.

Bins or classes: These are the subintervals of equal length that are used in constructing a histogram.

Frequency or count: the number of observational units in a subinterval.

Relative frequency or proportion: the fraction of observational units in a subinterval

Sample Size: the number of observational units studied in a sample

Important Points To Remember

Consistency refers to the spread in a distribution. Consistent means data are clustered in a small range

The best way to begin the analysis of data is by creating a graph, called distribution.

Histograms are used to depict data from quantitative variables.

Right skewed dot plots have most of their data points to the left.

The median of a set of observations not always be one of the observations within that set.

If an observation is further from the Q1 or **Q3** than 1.5 times the interquartile range, it is an outlier.

The median of a distribution is resistant to outliers, but the mean is not.

The measures of center apply to quantitative variables only

Both mean and median are measures of center, center is a property of the distribution.

In a symmetric, mound-shaped distribution, the mean, median and mode are equal.

Measures of spread apply to quantitative data only.

The center and the spread are two different properties of a distribution and there is no relation between the value of their measures. SD can be smaller or larger than the mean, IQR can be smaller or larger than the median

The lower quartile is the **median** of the lower half of the sample.

In any boxplot, the length of the whiskers is not always equal to $1.5 \times \text{IQR}$.

In left-skewed distributions, the distance from the smallest observation to Q1 exceeds the distance from Q3 to the largest observation.

When there is an odd number of observations, you do not include the actual median in either group.

If 10 is added to every value of a data set, the mean increases by 10.

If 10 is added to the maximum value of a data set, the median remains the same.

Two distributions have identical boxplots, the distributions may not have the same dotplots

Bar Graph: (used for categorical data)

Horizontal axis used to list the categories and labeled with the name of the variable of interest.

Bars have to be separated from each other and have different colors or patterns.

Bar height represents the frequency of the category of the bar.

Vertical axis show the frequency.

Mode can be recognized in a certain interval (bin)

The interval (bin) width has no meaning

Dot plot: (used for quantitative Data)

Used for small dataset

It has all the data values.

Each dot represents one observational unit data value

Sample size can be recognized.

Minimum and maximum data values can be recognized.

The frequency of any data value can be recognized.

Mode can be recognize.

Data can be used to calculate Mean, Median, Q1 and Q3.

Can show symmetry, skewness, gaps, and clusters.

Stem plot: (used for quantitative data) (It can be Stem plot, split stem plot or side by side stemplot)

Used for small dataset and each to construct by hand

It has all the data values.

Each number represents one observational unit data value

sample size can be recognized.

Minimum and maximum data values can be recognized.

The frequency of any data value can be recognized.

Mode can be recognized

Data can be used to calculate Mean, Median, Q1 and Q3.

Can show symmetry, skewness, gaps, and clusters.

If you given the median, you should be able to figure out the stems key

Boxplot: (used for quantitative data)

Used with large dataset.

Actual data values cannot be recognized.

You cannot know how many data points are in each quarter

Sample size is unknown.

Mean is unknown.

Five number summary can be recognized (Minimum, Q1, Median, Q3, Maximum)

Percentage of the data values in each part of the box can be recognized (25%, 50%, 75%, 100%), but not at single value.

Can show symmetry, skewness and outliers.

Histogram: (used for quantitative data)

Used with large dataset.

Actual data values cannot be recognized.

Sample size can be recognized.

Mean is unknown.

Median and mode can be guessed in which bins they are located.

Bin width affects the shape of the data, so you can get different histograms for the same dataset.

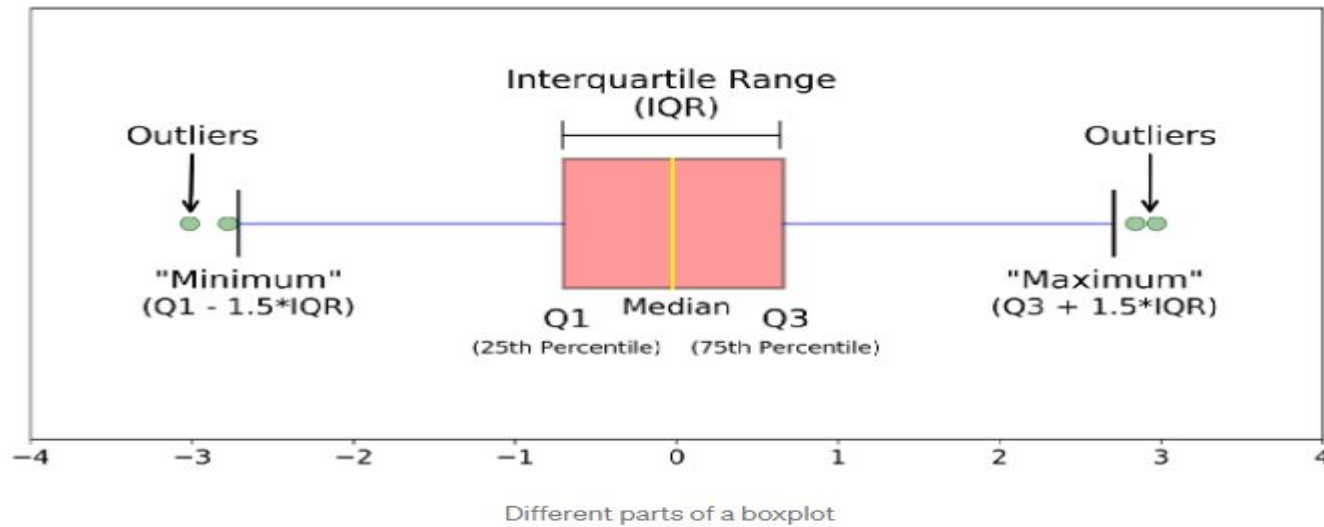
Number of data points (frequency) in each bin can be recognized but not at a single value because the data points are scattered inside the bin

Can show symmetry, skewness, gaps, and clusters.

Bars are constructed whose heights correspond to the frequency in each subinterval.

Bins: these are the subintervals of equal length that are used in constructing a histogram.

Increasing number of bins will show the shape but it does not mean more variability



Median (Q2/50th Percentile): the middle value of the dataset.

First quartile (Q1/25th Percentile): the middle number between the smallest number (not the “minimum”) and the median of the dataset.

Third quartile (Q3/75th Percentile): the middle value between the median and the highest value (not the “maximum”) of the dataset.

InterQuartile Range (IQR): 25th to the 75th percentile.

Whiskers (shown in blue)

Outliers (shown as green circles)

“Maximum”: $Q3 + 1.5 \cdot IQR$

“Minimum”: $Q1 - 1.5 \cdot IQR$

Outlier Calculations:

$$\text{IQR} = Q3 - Q1$$

$$1.5 \text{ IQR}$$

$$\text{LF} = Q1 - 1.5 \text{ IQR}$$

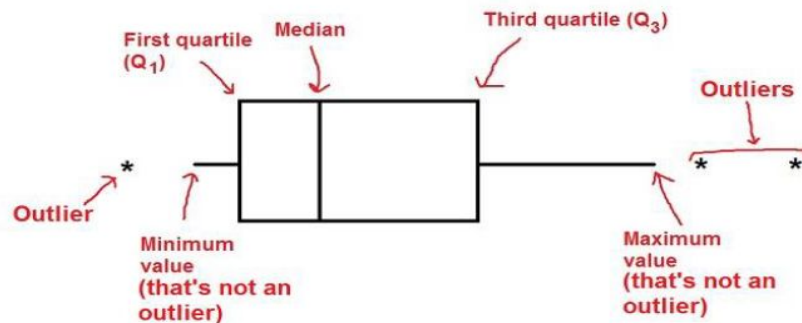
$$\text{UF} = Q3 + 1.5 \text{ IQR}$$

Outliers exist if the value is below your lower fence and above your upper fence.

Creating Boxplots and Identifying Outliers

A **boxplot** is a great way of quickly telling the spread and distribution of the data. We use the five-number summary (min, Q1, med, Q3, max) to create it. The steps are:

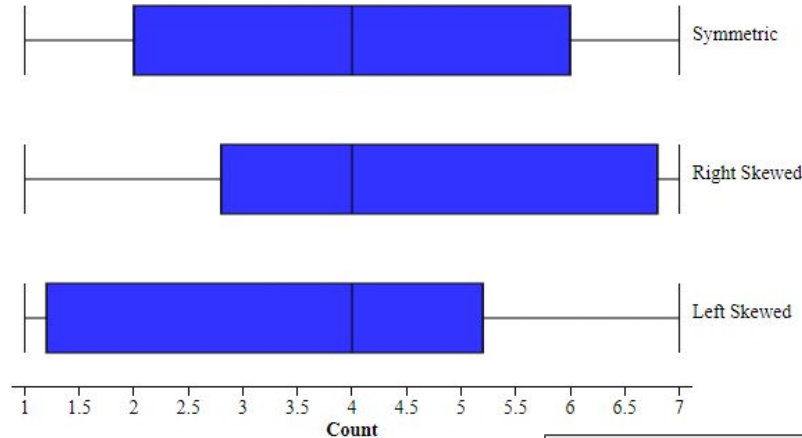
1. Draw a box from Q1 to Q3—that's the middle 50% of the data.
2. A vertical line inside the box marks the median.
3. Lines extend from the box out to the smallest and largest observations that are not outliers.



We will always include the outliers (as shown here) as separate points/marks from the line that extends from the box to the last value that is not an outlier in either direction. The textbook refers to this as a “(modified) boxplot” but it is the only type we will use in this class.

If the Median and IQR are given you can not figure out Q1 and Q3, so you can not figure where is the middle 50% of your data located exactly.

These distributions have the same median and IQR but different Q1 and Q3



Summary Statistics

Group Name	n	mean	SD	min	Q ₁	med	Q ₃	max
1: Symmetric	7	4	2.16	1	2	4	6	7
2: Right Skewed	7	4.229	2.195	1	2.8	4	6.8	7
3: Left Skewed	7	3.771	2.195	1	1.2	4	5.2	7

Five -Number
Summary

In boxplots you cannot know the percentage of the data in any arbitrary interval.

You can only know the percentage of the data in these exact intervals:

[Minimum, Q₁] is 25%

[Q₁, median] is 25%

[median, Q₃] is 25%

[Q₃, Maximum] is 25%

$$\text{range} = \text{max} - \text{min}$$

The range and IQR have to be reported as a single numbers, and they can not be used with categorical data.

These data reveal that the mean is close to the median with symmetric distributions, while the mean is greater than the median with distributions that are skewed to the right and the mean is less than the median with distributions that are skewed to the left. In other words, when one tail of the distribution is longer, the mean follows the tail.



Watch Out

These are not hard and fast rules. Do not look at the relative positions of the mean and median to decide about the shape of the distribution. Rather, look at the shape of the distribution to predict how the mean and median will compare to each other.

Mean = Median	---->	Symmetric
Mean > Median	---->	Skewed to the right
Mean < Median	---->	Skewed to the left

Hint: When comparing the Mean to The Median:

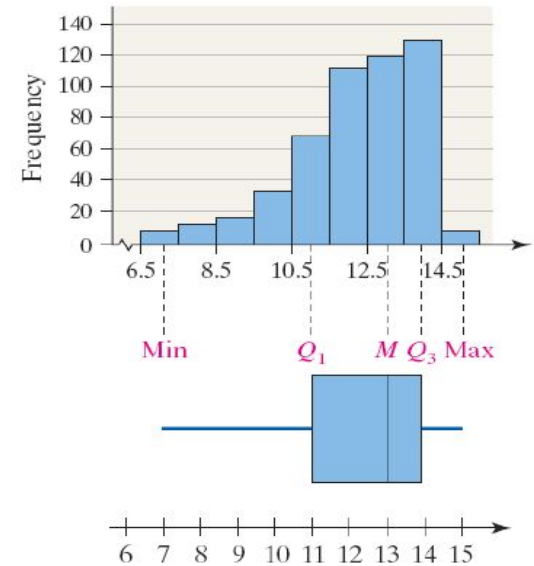
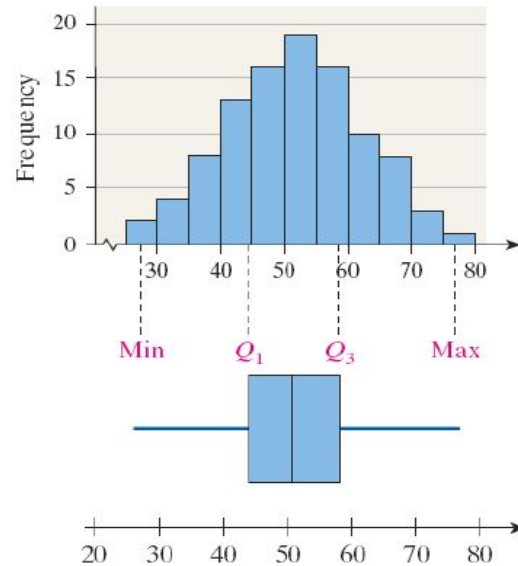
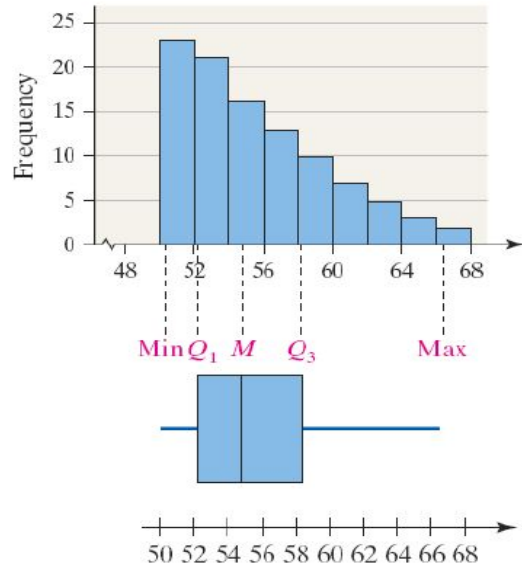
the **<** or **>** symbols points to the skewed side

When one tail is longer, the mean follows the tail

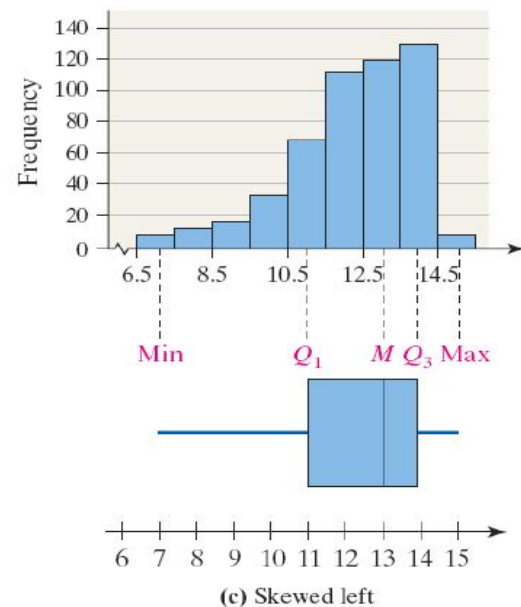
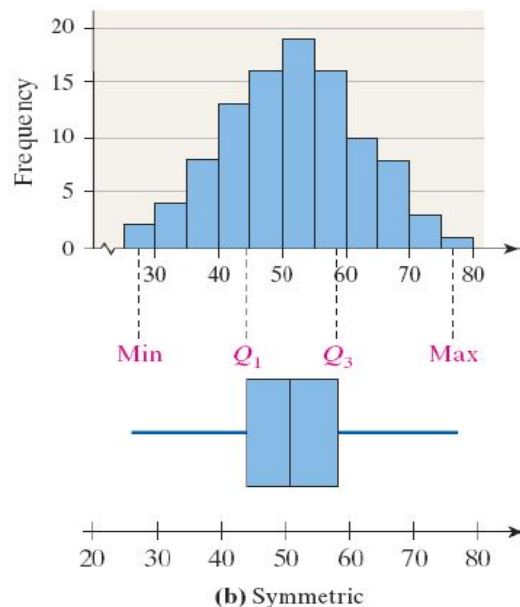
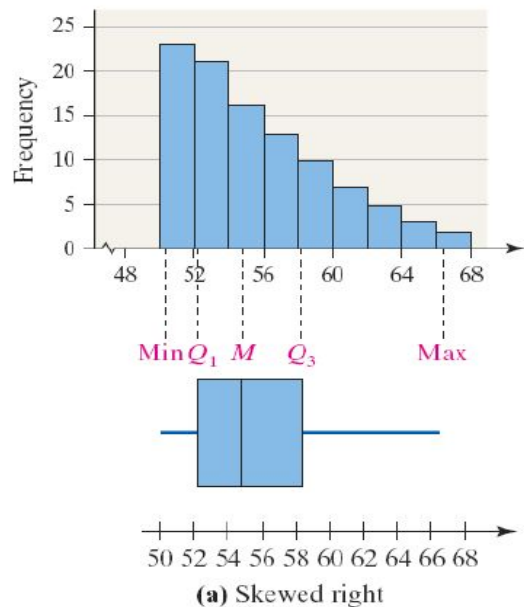
If the data is given in a table that lists the frequency of each data point, the side of the table that has data points with less frequencies is where the tail is located and the side with high frequency is where the peak is located.

If the the higher frequencies in the middle, so the shape may be symmetric or roughly symmetric

Use a boxplot and quartiles to describe the shape of a distribution.

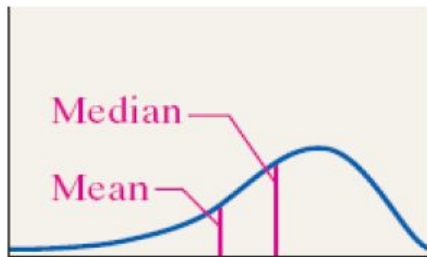


Use a boxplot and quartiles to describe the shape of a distribution.



Relation Between the Mean, Median, and Distribution Shape

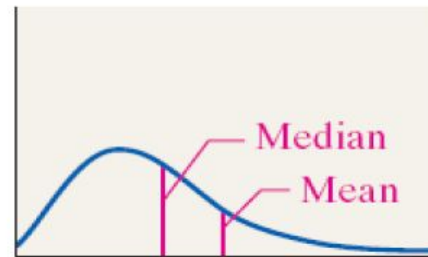
Distribution Shape	Mean versus Median
Skewed left	Mean substantially smaller than median
Symmetric	Mean roughly equal to median
Skewed right	Mean substantially larger than median



(a) Skewed Left
 $\text{Mean} < \text{Median}$



(b) Symmetric
 $\text{Mean} = \text{Median}$



(c) Skewed Right
 $\text{Mean} > \text{Median}$

Shape of the distribution: to be sure about the shape you have look over the relation between the mean and the median in summary Statistics.

Calculating IQR

Calculation of the outliers and figuring the outliers

Measures used for central tendency or spread according to the shape of the distribution

For symmetric shape and no outliers use mean and standard deviation

For skewed shape or if there are outliers use median and IQR

Describing the center and spread of a distribution of one group

The mean/median is **value** and measuring units and The SD/IQR is **value** and measuring units. A Typical observational unit and variable name tends to be **value** and measuring units and there is high/low variability.

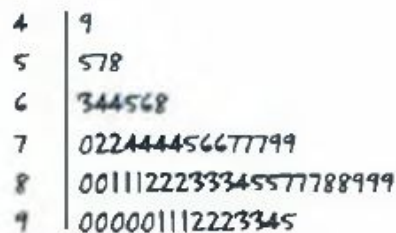
When comparing the center and spread of a distribution of two groups:

The mean/median for group 1 name is **value 1** and measuring units while the mean/median for group 2 name is **value 2** and measuring units. The SD/IQR for group 1 name is **value 1** and measuring units while the SD/IQR for group 2 name is **value 2** and measuring units.

A typical **Observational unit** from group 1 name tends to have higher/lower (variable name) than group 2 name and group 1 has more/less variability than group 2.

Exercise 7-21: Exam Scores

The following stemplot displays the scores of 62 students who took the first exam in a statistics course:

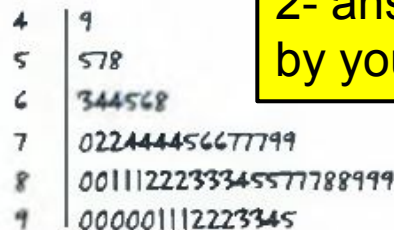


- 1- Label the stemplot
- 2- answer the questions
- 3- Describe the shape

- a. How many students received a score of 77 on the exam?
- b. How many and what proportion of students scored 90 or greater?
- c. How many and what proportion of students scored less than 70?
- d. Which score appears most often?
- e. There are only two values between 75 and 95 for which no one obtained those particular scores. What are the two values?

Exercise 7-21: Exam Scores

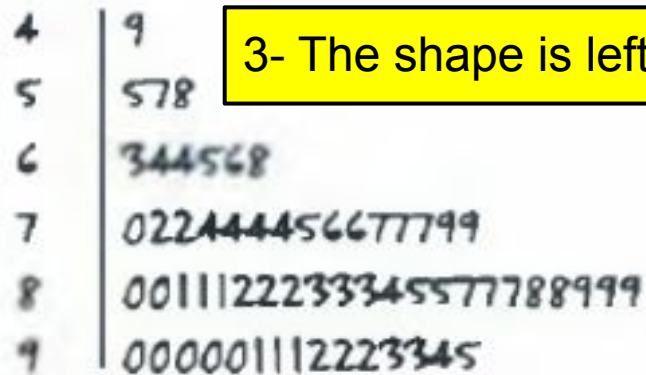
The following stemplot displays the scores of 62 students who took the first exam in a statistics course:



2- answer the questions by yourself

- How many students received a score of 77 on the exam?
- How many and what proportion of students scored 90 or greater?
- How many and what proportion of students scored less than 70?
- Which score appears most often?
- There are only two values between 75 and 95 for which no one obtained those particular scores. What are the two values?

1- Label the stemplot



3- The shape is left skewed

The Scores of 62 Students who Took the First Exam in Statistics

KEY: 4|9 = 49

Side-by-Side Stemplot

2009 class		2010 class
9, 8, 8, 8, 7, 6, 6, 6, 6	7	0, 0, 0, 1, 2, 4, 4, 4, 4, 5, 6, 6, 7, 8, 9
8, 5, 5, 3, 3, 3, 2, 2, 0, 0	8	0, 1, 2, 2, 2, 3, 4, 5, 5, 6, 7
5, 1	9	3, 8
	10	0

Compare the shapes of the 2010 class and 2009 class distributions

Split Stemplot

11	44
11	66778
12	0134
12	666778888
13	0000001111444
13	7788999
14	0044
14	567
15	11
15	
16	0

Percentage of State's Residents
Aged between 25 and 34

KEY: 12|1 = 12.1%

Side-by-Side Stemplot

2009 class		2010 class
9, 8, 8, 8, 7, 6, 6, 6, 6	7	0, 0, 0, 1, 2, 4, 4, 4, 4, 5, 6, 6, 7, 8, 9
8, 5, 5, 3, 3, 3, 2, 2, 0, 0	8	0, 1, 2, 2, 2, 3, 4, 5, 5, 6, 7
5, 1	9	3, 8
	10	0

Compare the shapes of the 2010 class and 2009 class distributions:

Both the distributions of the 2010 class and 2009 class are right skewed

Split Stemplot

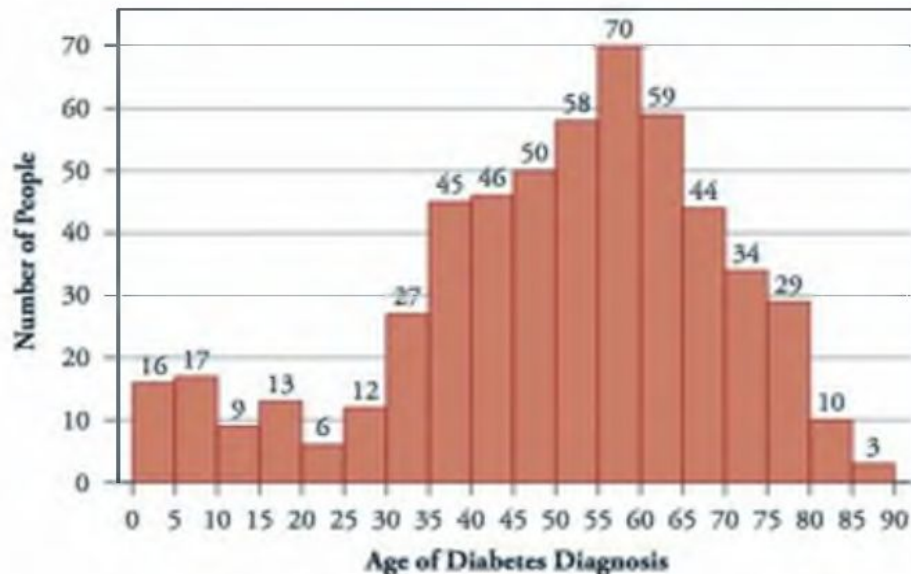
11	44
11	66778
12	0134
12	666778888
13	0000001111444
13	7788999
14	0044
14	567
15	11
15	
16	0

Percentage of State's Residents
Aged between 25 and 34

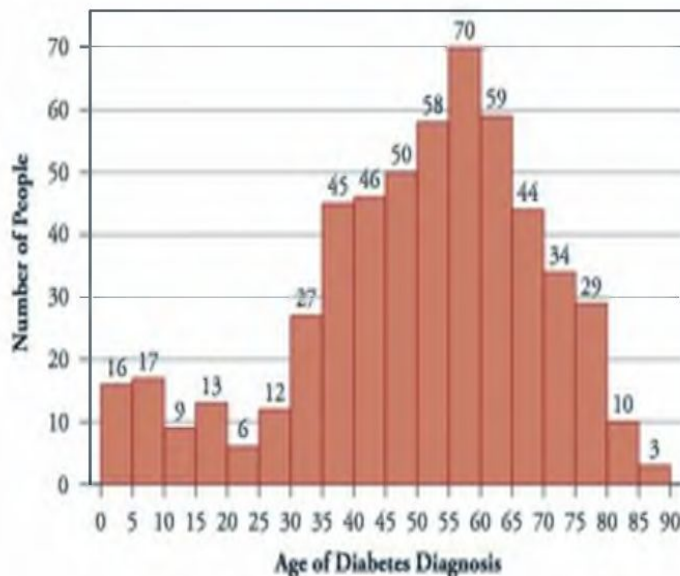
KEY: 12|1 = 12.1%

Explain the data that you can get from this histogram

The following histogram displays the distribution of ages at which the 548 survey subjects were diagnosed with diabetes. Note that the endpoints of the subintervals are reported. For example, the first subinterval indicates that 16 people were diagnosed with diabetes before age 5, and the second subinterval represents the 17 people who were at least 5 years of age but younger than 10 years of age when they were diagnosed with diabetes. (A person aged 5 years would appear in the 5–10 interval, not the 0–5 interval.)



The following histogram displays the distribution of ages at which the 548 survey subjects were diagnosed with diabetes. Note that the endpoints of the subintervals are reported. For example, the first subinterval indicates that 16 people were diagnosed with diabetes before age 5, and the second subinterval represents the 17 people who were at least 5 years of age but younger than 10 years of age when they were diagnosed with diabetes. (A person aged 5 years would appear in the 5–10 interval, not the 0–5 interval.)



Explain the data that you can get from this histogram

70 people diagnosed with diabetes at age between 55 and 59 years old

The probability of a randomly select one diagnosed with diabetes at age between 55 and 59 years old is $70/548$

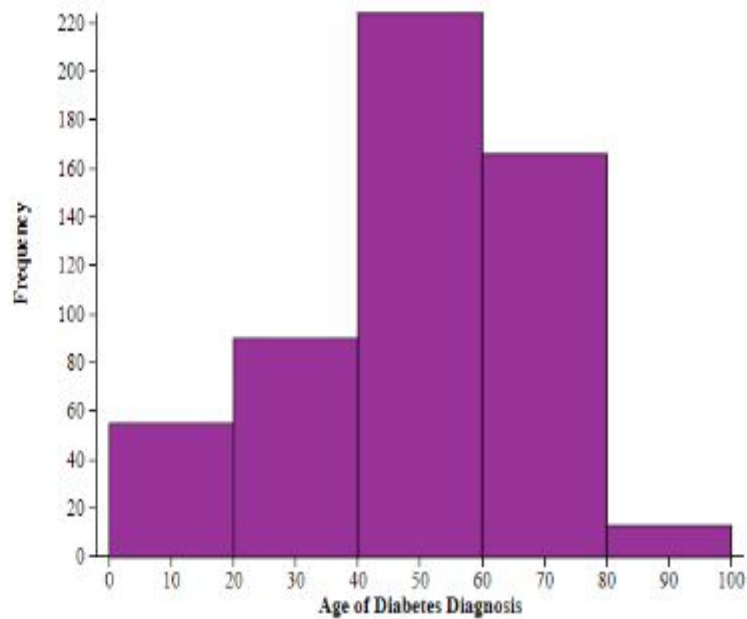
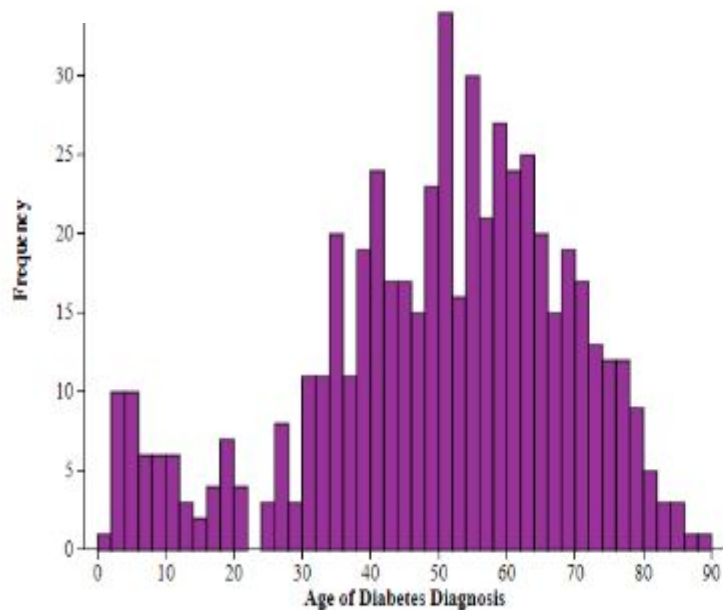
55 people diagnosed with diabetes at age less that 20 years old

The shape is skewed to the left

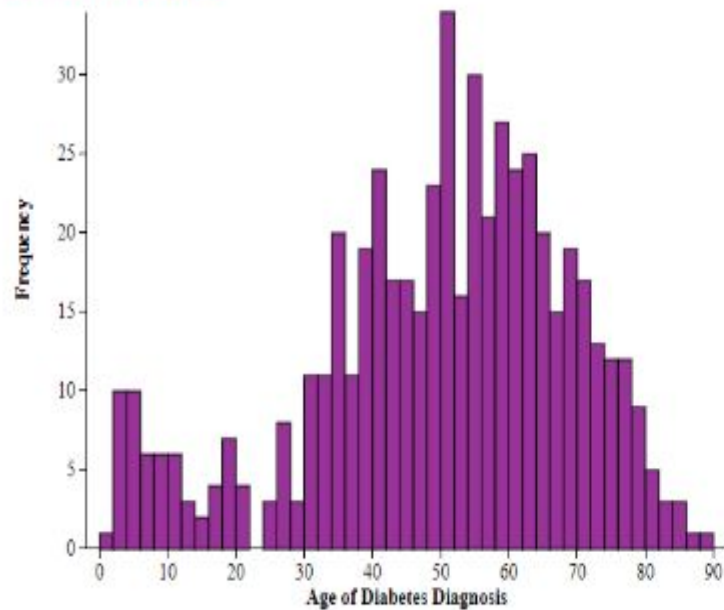
The median is located inside 50-55 bin

I can not find how many people diagnosed with diabetes at an exact age like 50 or 65 years old

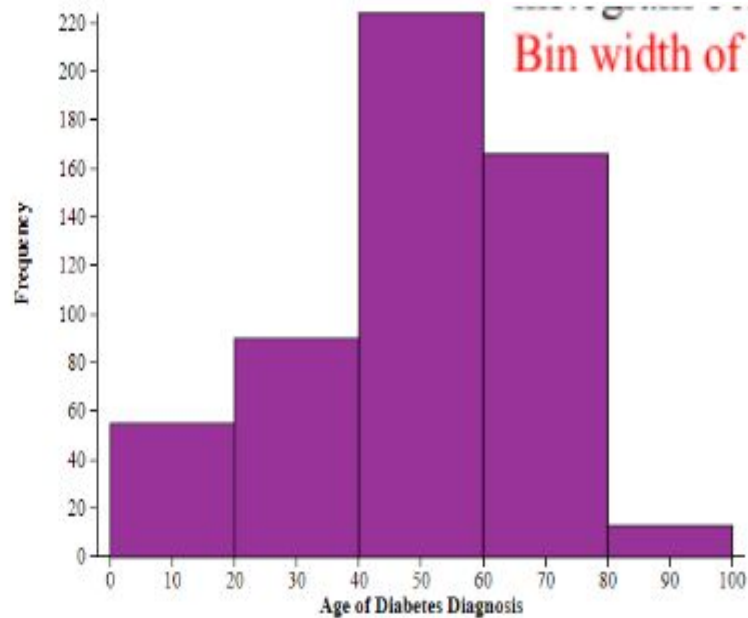
What is the Bin width of each histogram?



Bin width of 2:

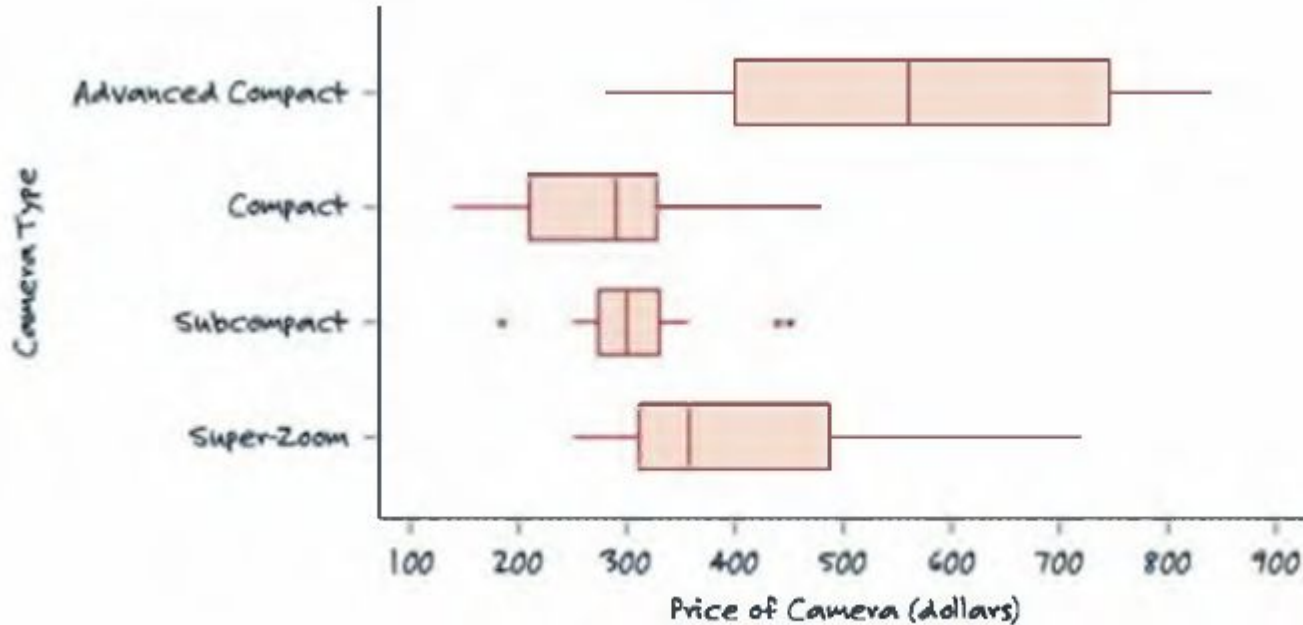


Bin width of 20:



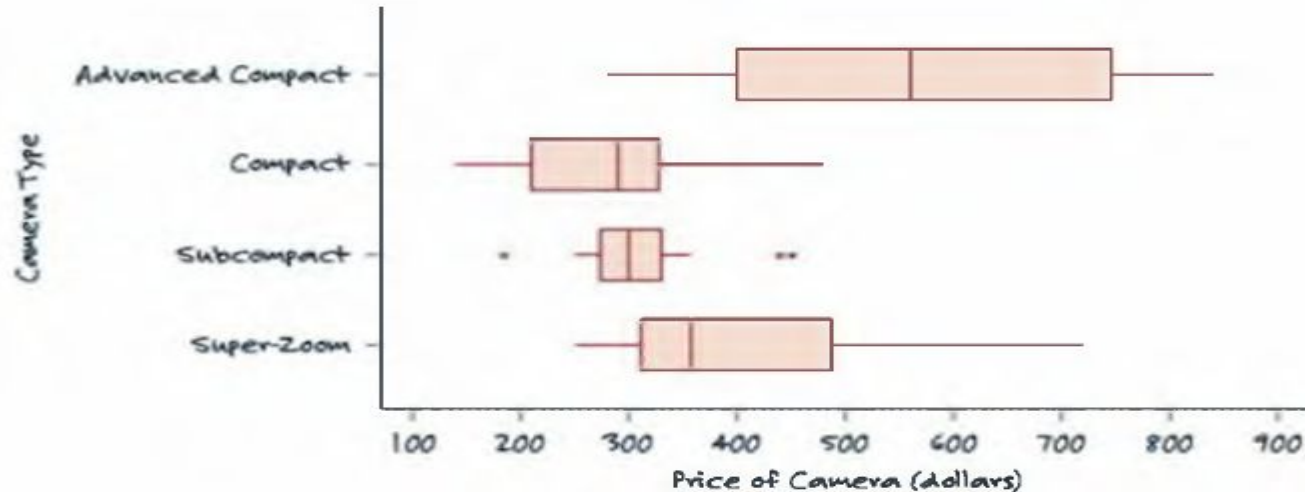
1- Describe the center and the spread of advanced compact cameras

2- Compare the distribution of the compact and super zoom cameras using the measures of center and spread



1- Describe the center and the spread of advanced compact cameras

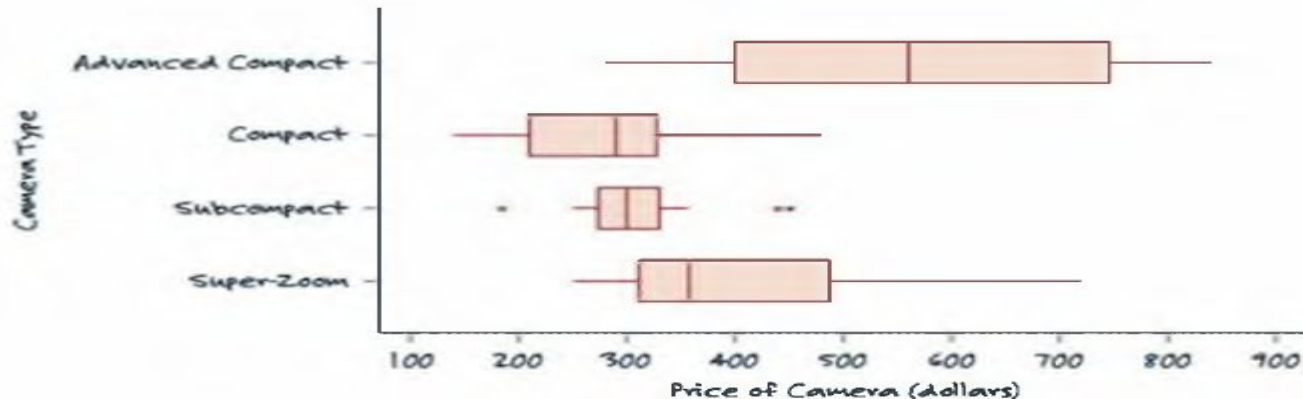
The median is **580** dollars and The IQR is **350** dollars. A Typical price of an advanced compact camera tends to be **580 dollars** and there is high variability.



2- Compare the distribution of the compact and super zoom cameras using the measures of center and spread

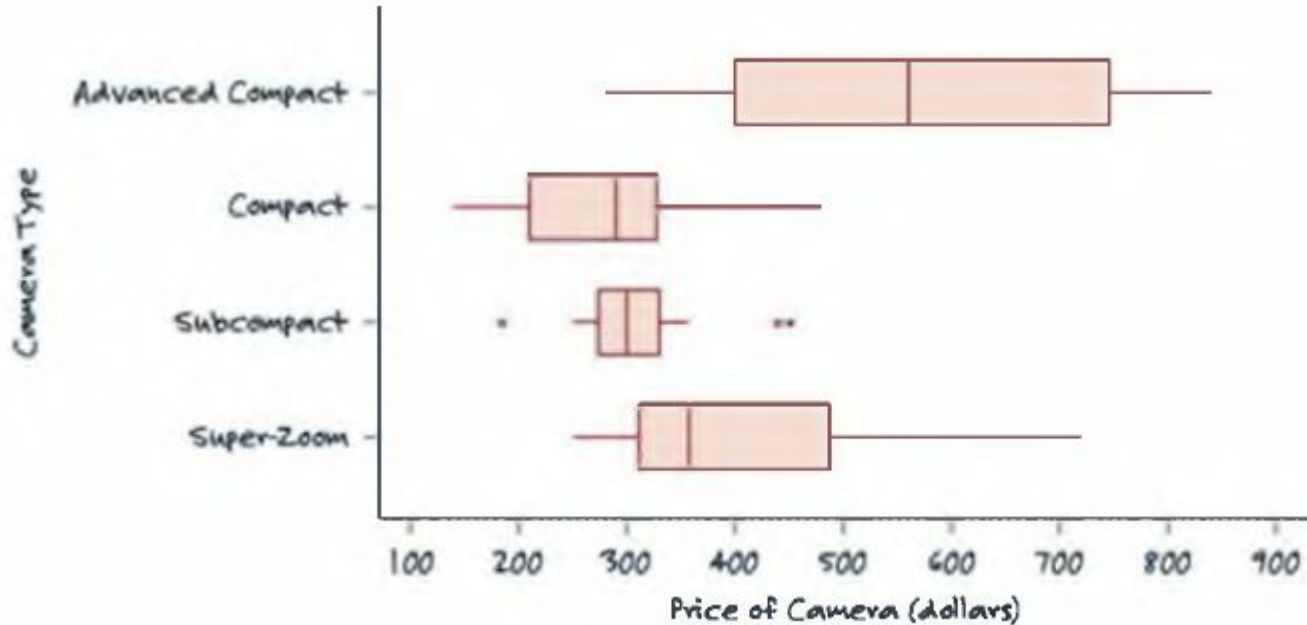
The median for compact cameras is **290 dollars** while the median for super zoom cameras is **340 dollars**. The IQR for compact cameras is **130 dollars** while the IQR for super zoom cameras is **280 dollars**.

A typical **compact camera** tends to have lower price than a typical super zoom camera, and the prices of compact camera have less variability than super zoom cameras.



1-Identify the observational units and the variables

2-Write a research question



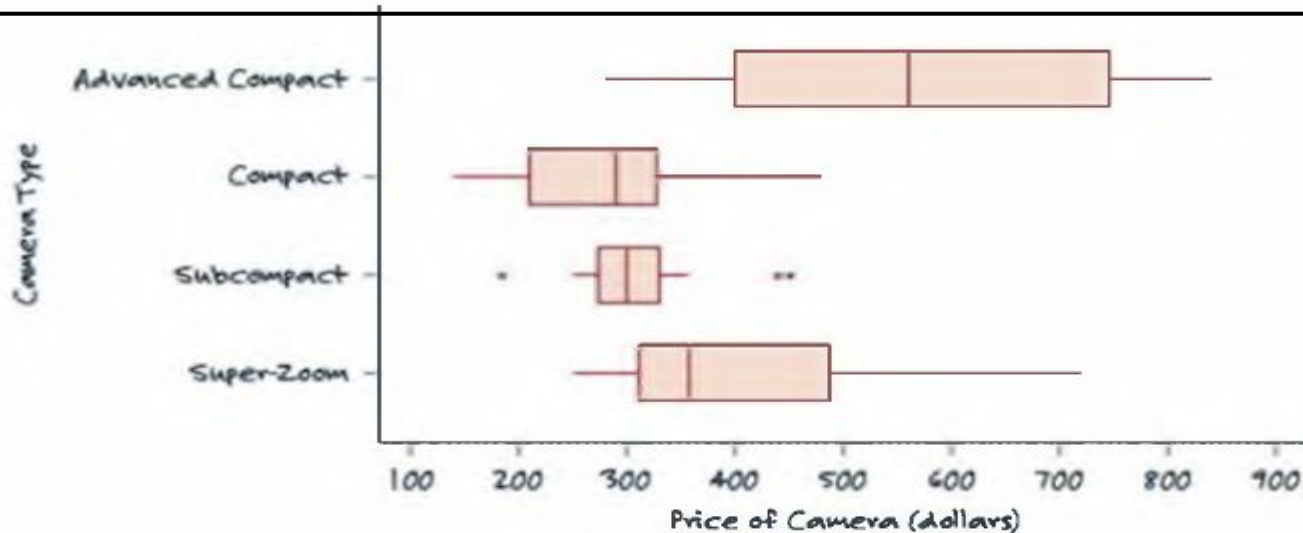
1-Identify the observational units and the variables

Observational Units: Cameras

Quantitative Variable: Price of the camera

Categorical Variables: Camera type

2-Write a research question: Which camera type tends to have lower price?
What is the typical price of subcompact camera?



Check for understanding

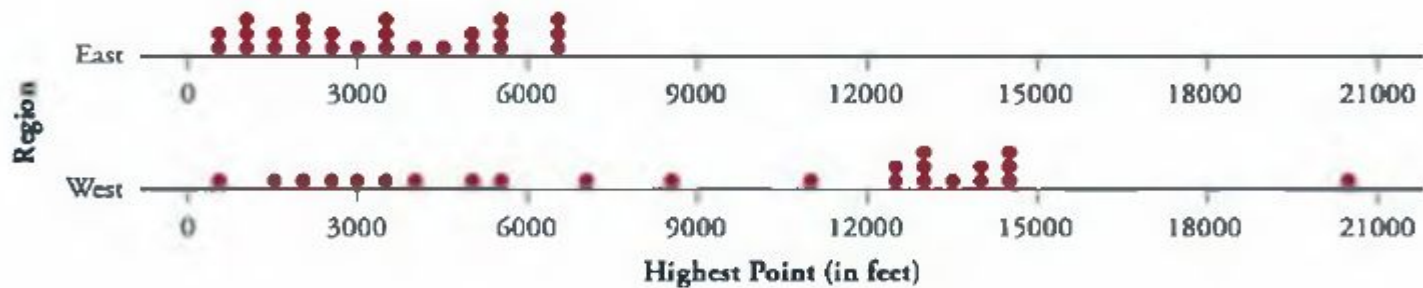


Figure Ex. 2-15

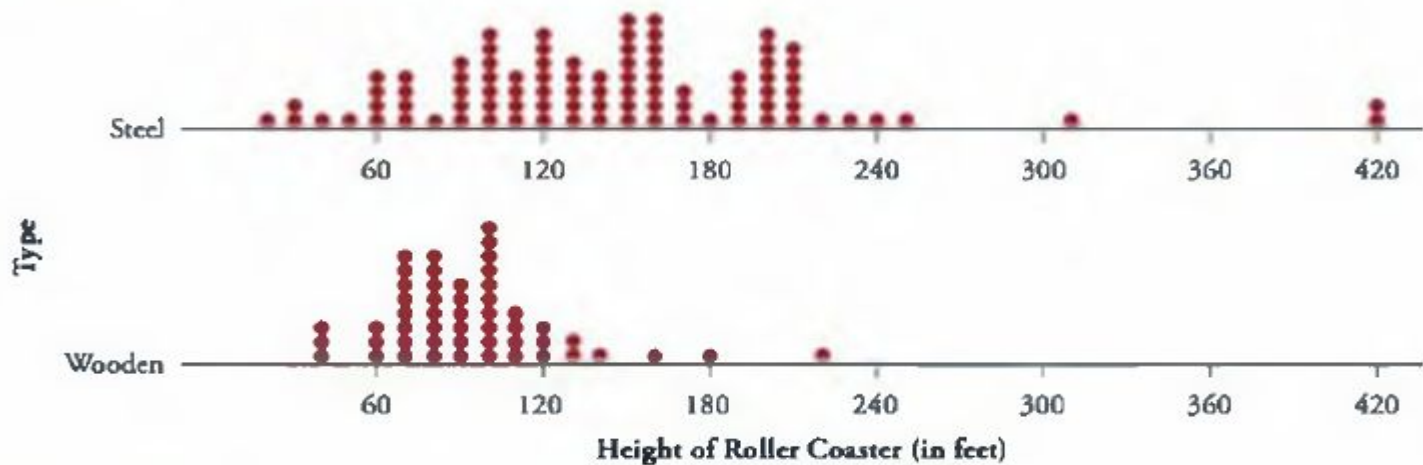


Figure Ex. 2-17

10. What is the arithmetic average and how is it found? **It is found by adding up the values for each of the observational units and dividing by the number of values. It is the balance point of the distribution.**
11. What is the value of the middle observation and how is it found? **Median. Once the n values have been arranged in numerical order, the median of an odd number of data values is located in position $(n + 1)/2$. The median of an even number of values is defined to be the arithmetic (mean) of the middle two values, on either side of position $(n + 1)/2$.**
12. Can mean and median be applied to categorical variables? **No.**
-
13. What measure appears most often in a distribution of data? **Mode**
14. Does a mode apply to categorical data? **Yes.**
15. What type of measure is relatively unaffected by the presence of outliers in a distribution? **a resistant measure**
16. Is the mean or median resistant? **The median is resistant.**

Watch Out!

Turn to page 160 and answer the questions based on the Watch Out.

18. Is center a property or measure? **Property. Mean and median are measures of center.**
19. What two ways do we measure center? **Mean and median.**
Which is preferable? **Neither; both have their own properties and strengths.**
20. What features are not indicated by center? **Spread, shape, clusters, and outliers.**

Watch Out!

Answer the following questions from the Watch Out on page 202.

1. Do you know how many data points are in each quarter of a boxplot?
No, unless you know the sample size.
2. Do all technologies construct boxplots in the same way? **No**

Watch Out!

Turn to pages 137-138 and answer the questions from the watch outs.

1. Does a bar graph or histogram display quantitative data?
Histogram.
2. What type of scale is on the horizontal axis of a histogram?
Numerical.
3. What are the subintervals of a histogram called? **Bins.**
Is there only one choice for these intervals? **No**

Topic 5

Anecdotal evidence results from situations that come to mind easily and is of little value in scientific research. Much of the practice of statistics involves designing studies and collecting data so that people do not have to rely on anecdotal evidence.

You have to do an experiment using random sampling to test any claim based on anecdotal evidence.

An **experiment** is a study in which the experimenter *actively imposes* the **treatment** (explanatory variable group) on the subjects. Ideally, the groups of subjects are identical in all respects other than the explanatory variable so that the researcher can then see the explanatory variable's direct effects on the response variable.

One way to avoid the effect of confounding variables and balance them between groups is "Random Assignment"

Random assignment is the preferred method of assigning subjects to treatments (explanatory variable groups) in an experiment: Each subject has an equal chance of being assigned to any of the treatment groups. Such a study is called a **randomized comparative experiment**.

Randomly Selected: This process takes place when you randomly select people or things to be in your sample, So your sample will represent the population of interest

Randomly Assigned: This process takes place when you randomly assign people/things to be in one of the two study groups (Treatment group or Placebo group). So two groups can be similar except for the treatment received

Aspects of Experimental Design are:

Randomization

Control group

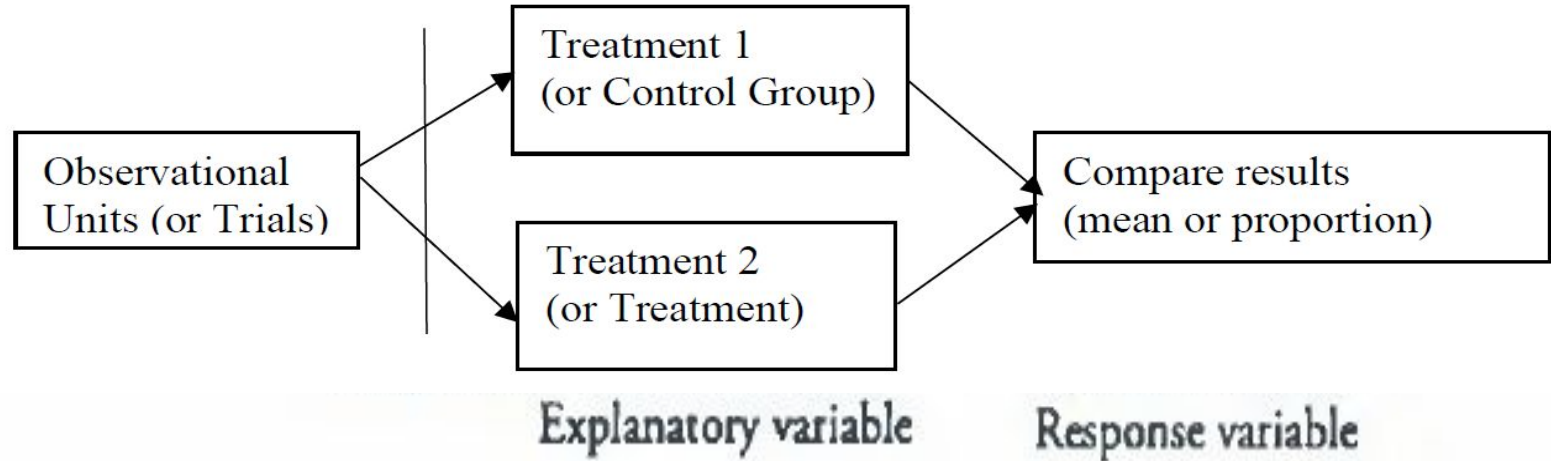
Treatment group

Blinding (Subject unaware)

Double Blinding (evaluator and Subject unaware)

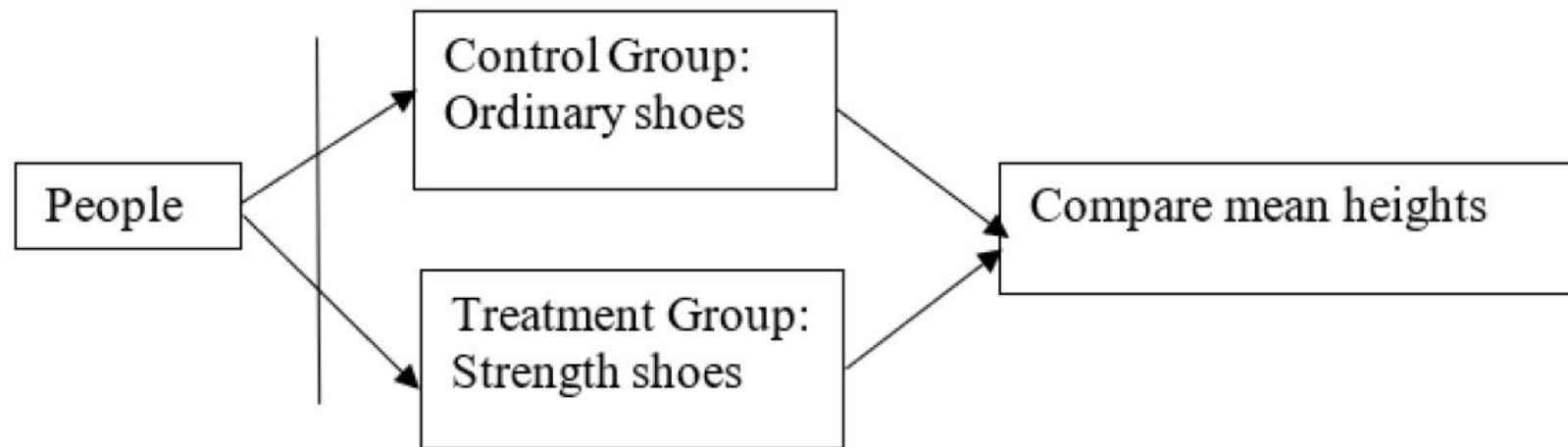
From a well designed experiment, you can have a cause and effect relationship

The design of an experiment is often visualized in a simple graphic:



where we can use any number of groups for the explanatory variable.

The experimental design diagram for the strength shoes experiment in Activity 5-2 would look like:



Experimenters try to assign subjects to groups so that lurking and potentially confounding variables tend to balance out between the two groups. These activities demonstrate that **random assignment** generally achieves the goal of creating treatment groups that are similar in all respects except for the treatment imposed. Thus, if the groups turn out to differ substantially on the response variable, you can draw a cause-and-effect conclusion between the explanatory and response variables. As you can see, a randomized comparative experiment has the potential to establish a cause-and-effect relationship between two variables.

Comparison groups are especially important in medical studies because subjects often **respond positively** simply to being given a treatment, whether or not the treatment is actually effective. This phenomenon is known as the **placebo effect**. Experimenters control for this by administering a **placebo** (a sugar pill with no active ingredient) to subjects in a control group or by comparing a new treatment against a standard treatment. Naturally, **subjects must be blind** as to which treatment they actually receive or else that knowledge could affect their responses. When possible, experiments should be **double-blind** so that the **person evaluating the subjects is also unaware** of which subjects receive which treatment. In this way, the evaluator's judgment is not influenced (consciously or subconsciously) by any hidden biases.

Good Stats: The Four Principles of Experimental Design

The term **experimental design** refers to a plan for assigning experimental units to treatment conditions. There are four key principles of experimental design:

- **Comparison.** Use a design that compares two or more treatments.
- **Random assignment.** Use chance to assign experimental units to treatments.
 - Doing so helps create roughly equivalent groups of experimental units by balancing the effects of other variables among treatment groups.
 - This can include completely randomized designs or randomized block design which are explained below.
- **Control.** Keep other variables the same for all groups, especially variables that are likely to affect the response variable.
 - Control helps avoid confounding and reduces variability in the response variable.
- **Replication.** Use enough experimental units in each group so that any differences in the effects of the treatments can be distinguished from chance differences between groups.

Watch Out after Self-Check, Activity 5-6:

1. Random **sampling** aims to produce a **sample** that is representative of the population. Random **sampling** eliminates **bias**.
2. Random **assignment** aims to produce **treatment groups** that are similar in all respects except for the treatment imposed. Random **assignment** eliminates **confounding**.

Some useful definitions to remember and habits to develop from this topic are

- An **experiment** is distinguished from an observational study by the experimenters deliberately imposing the explanatory variable on the subjects, rather than passively observing and recording information.
- A **randomized comparative experiment** is the gold standard for establishing a cause-and-effect relationship between two variables. By randomly assigning subjects to treatment groups, you guard against the possibility that the groups differ systematically in another way (i.e., you guard against confounding).
- **Random assignment**, or randomization, is a different use of randomness than is random sampling. Although random assignment allows you to potentially draw cause-and-effect conclusions, be cautious about generalizing the results of randomized experiments to a larger population unless the subjects were randomly sampled from the population of interest.
- **Blindness** refers to the subjects in an experiment being unaware of which treatment they receive. **Double-blindness** refers to the evaluators (not the researchers so much as those determining the response variable outcome) also being unaware of which group received which treatment.
- The scope of conclusions that you can draw from a study depends primarily on how the data were collected. Random *sampling* allows for **generalizing** from a sample to the larger population. Random *assignment* plus a

statistically significant difference among the groups allows for drawing a **cause-and-effect** conclusion between the explanatory and response variables. When describing study conclusions, address generalization, causation, and significance separately.

You should be able to

- Recognize the advantage of systematically collected data over anecdotal evidence. (Activity 5-1)
- Classify a study as an experiment or an observational study. (Activity 5-1)
- Explain the goals and usefulness of using random assignment in an experiment. (Activities 5-1, 5-2, 5-3)
- Implement random assignment to assign subjects to treatments in an experiment. (Activities 5-1, 5-2)
- Comment on other desirable features of a well-designed experiment, such as comparison, blindness, and double-blindness. (Activities 5-4, 5-5)
- Identify when, and explain why, a cause-and-effect conclusion can be drawn from the results of an experiment. (Activity 5-3)
- Recognize the distinction between random sampling and random assignment, and explain how the scope of conclusion that can be drawn from a study depends on which type(s) of randomness were used in the process of collecting data. (Activity 5-5)

Topic 5: Designing Experiments

- Four Principles of Experimental Design

1. Compare two (or more) treatments

2. Random Assignment

- Randomly assign subjects to treatment groups
- Goal: to create groups similar in all aspects except for treatment
- Why use it?
 - Establish “cause-and-effect” relationship between explanatory and response variable
- Different than random sampling
 - How? They have different objectives
 - Random sampling: allows generalizing results to the population since the population is adequately represented in the sample
 - Random assignment: allows for determining cause-and-effect conclusion between explanatory and response variables

Topic 5: Designing Experiments

- Four Principles of Experimental Design

3. Control

- Keep everything the same between the groups as much as possible
- Possibly administer placebo to one group
- Blind: Subjects do not know their assigned group
 - Placebo group or Treatment group?
- Double Blind: Neither researchers or subjects know which group is receiving placebo vs. treatment
 - Why use double blind?

4. Replication

- Use enough experimental units so that differences in treatments can be attributed to the treatments & not the differences in subjects

Biased samples: Volunteers, convenient, Response , non-response, undercover

Usually volunteer desire to succeed or treated will be a confounding variable.

Voluntary Response Bias refers to samples collected in such a way that members of the population decide for themselves whether or not to participate in the study, **it often represents people with strong opinion**

Convenience Sample Bias samples that are easily accessible, **it often leads to undercoverage bias**

Response bias the behavior of the respondent or interviewer or the wording of the question can cause **response bias** in sample results

Nonresponse Bias members of the population **did not have** the opportunity **to respond**

Is increasing sample size will decrease the bias !!!!!

Use Unit 3 Vocab Matching file(It was distributed in class)

- Print it out
- Cut it
- Match the different vocab words with their definitions (Match page 1 with 3 and Match page 2 with 4)

Note: When you do the matching it is a good practice to read the definition first and then match it with the right vocab word

Simple Random Sampling (SRS) when every possible sample of size n has **an equal chance of being the sample ultimately selected**

Random Assignment: to **balance the effect of the confounding variables**

Random Assignment it creates treatment **groups that are similar in all respects except for the treatment imposed** so that lurking and potentially confounding variables tend to balance out between the two groups.

Observational Study when the investigator **passively** observes and records information on observational units

Experiment (Randomized Comparative Experiment): A study in which the experimenters actively **impose the treatment** on the subjects and can establish a cause and effect relationship

Sampling Variability the tendency of values of a sample statistic to **vary from sample to sample**

Precision (of a sample statistic) how much the values vary from sample to sample

Bias **tendency of samples to differ** from the corresponding population as the result of systematic exclusion of some part of the population

Sampling Bias systematic tendency of a sampling method to **overrepresent** some parts of the population

Blindness when the **subjects are not aware** which treatment, they actually receive

Double Blindness when **neither the person evaluating the subjects nor the subjects are aware** of which treatment they actually receive

Placebo a treatment with no active ingredient or benefit (such as a sugar pill)

Placebo Effect subjects often **respond positively** simply by being given a treatment

Anecdotal Evidence results from situations that **come to mind easily** and is of little value in scientific research